PDSS-Net: Phase-Dominant Structural Sensing and Symmetric Spectral Consensus for Remote Sensing Change Detection

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***Abstract*—Bitemporal change detection (CD) faces a persistent "same object, different appearance" dilemma, where phenological shifts entangle genuine structural evolution with seasonal stylistic variations. Conventional spatial methods struggle to decouple these factors, while existing frequency-domain approaches suffer from spectral geometric bias due to rigid Cartesian grids and metric asymmetry in feature interaction. To address these challenges, we propose the phase-dominant structural sensing network (PDSS-Net). Grounded in a frequency-domain disentanglement philosophy, PDSS-Net reformulates CD as extracting stable structural signals (phase) from unstable seasonal styles (amplitude). To implement this philosophy, we design two core components. First, we introduce the geometric-aware frequency alignment (GFA) module to rectify spectral geometric bias. Instead of applying rigid grid-based filtering, it aligns feature extraction with the inherent radial energy distribution of remote sensing scenes, thereby precisely capturing direction-dependent structural details. Second, we devise the bitemporal structure-guided frequency difference (BSFD) module to resolve interaction asymmetry. It maps bitemporal features into a unified style representation to neutralize seasonal discrepancies, while simultaneously safeguarding critical boundary topology to prevent the over-smoothing of genuine changes during harmonization. Extensive experiments demonstrate that PDSS-Net consistently achieves state-of-the-art performance across six challenging benchmarks, yielding intersection over union (IoU) scores of 84.61% on LEVIR-CD, 90.68% on Google, 59.80% on CLCD, 92.47% on DSIFN, 53.07% on SECOND, and 91.22% on BCDD. Code is available at: https://github.com/Zhongjx13/PDSSNet**

***Index Terms*—Deep learning, frequency domain, remote sensing change detection (RSCD), spatial domain.**

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I. Introduction

BITEMPORAL remote sensing change detection (RSCD) is a cornerstone of remote sensing interpretation but faces severe challenges from radiometric intraclass variance [1], [2]. In heterogeneous landscapes, long temporal intervals introduce significant phenological shifts (e.g., vegetation cycles) and illumination variations. This "same object, different appearance" phenomenon entangles genuine structural evolution with seasonal pseudo-changes, making accurate detection arduous and leading to pervasive false positives (FPs) [3], [4].

Over the past decade, deep learning has revolutionized the field of RSCD. Convolutional neural networks (CNNs) have demonstrated robust representational capabilities via hierarchical feature extraction [5], [6], while Transformer-based architectures have excelled in modeling long-range dependencies through self-attention mechanisms [7], [8]. Nevertheless, these mainstream approaches are intrinsically confined to feature aggregation within the spatial domain. In this domain, the geometric structure (semantic identity) and phenological style (imaging condition) of land covers exhibit a high degree of nonlinear coupling at the pixel level. Specifically, the inherent texture bias of CNNs predisposes them to overfit specific phenological states, often misinterpreting natural vegetation succession as genuine land-cover transitions [9].

Conversely, the intensity-based attention mechanisms in Transformers [7], [8], [10], [11], [12], [13] are highly sensitive to radiometric inconsistencies, where solar angle variations and shifting shadows disrupt the semantic correspondence between unchanged bitemporal objects. Consequently, these spatial methods struggle to distinguish structural remodeling from phenological evolution, leading to severe FPs in cross-seasonal detection. Fig. 1 visualizes the similarity heatmaps computed via standard cross-attention mechanisms on bitemporal images. In examples (a) and (b), the similarity metrics for semantic changes fail to exhibit a distinct contrast against the background, making them virtually indistinguishable. While example (c) correctly highlights the change region with low similarity values, it suffers from significant interference: the road background in the lower section also manifests low similarity due to imaging discrepancies. Consequently, the mechanism fails to separate these background variations from genuine structural changes.

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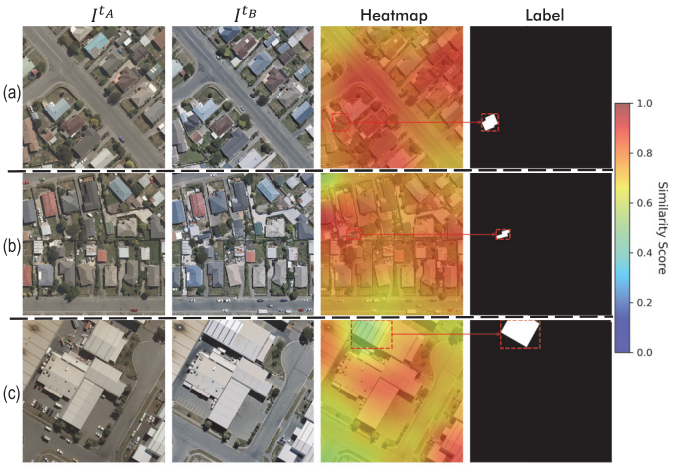


Fig. 1. Similarity heatmaps generated by standard cross attention. (a) and (b) Insufficient contrast between changes and the background. (c) Imaging discrepancies in the lower road background cause significant interference, appearing indistinguishable from the genuine change region.

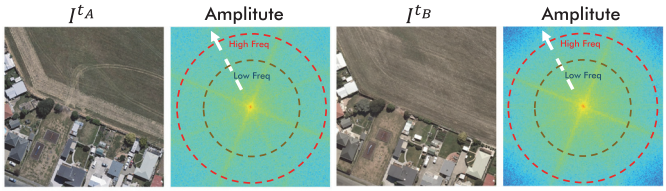


Fig. 2. Amplitude spectra of bitemporal inputs. The frequency components are distributed radially from the center, creating a concentric pattern.

To circumvent this spatial entanglement, recent studies have integrated frequency-domain learning as a complementary perspective [14], [15]. By leveraging the inherent decoupling properties of the Fourier Transform—where amplitude encodes style and phase encodes structure—these methods aim to enhance the robustness of spatial representations against phenological shifts [16], [17]. However, existing frequency-domain approaches in RSCD remain superficial, largely because they ignore the fundamental spatial properties of the frequency domain. As visually demonstrated in Fig. 2, the amplitude spectra of remote sensing images are inherently heterogeneous, with frequency components distributed radially from the center. Specifically, the low-frequency components representing seasonal styles are clustered at the center, while high-frequency structural edges of man-made objects (e.g., buildings and roads) manifest as direction-dependent energy ridges toward the periphery. Despite these distinct spectral signatures, current methods erroneously treat the spectrum as a homogeneous tensor. By forcing a standard, translation-invariant kernel to slide across the entire spectrum, they fail to apply location-adaptive processing to the center and the periphery. Consequently, these methods lose the capability to disentangle genuine structural evolution from phenological noise. Given this concentric topology, applying standard convolution kernels—which are defined on a rigid Cartesian grid—inevitably introduces an inherent geometric bias during feature extraction.

Critically, the prevailing paradigm of employing simple bitemporal feature subtraction to generate bitemporal change candidates is theoretically ill-posed [7], [18], [19], [20], [21]. In the real world, seasonal changes such as shifting sunlight act similar to a dimmer switch, which effectively scales the brightness up or down via multiplication rather than merely adding noise. Simple subtraction cannot handle this scaling effect. As a result, the model becomes confused and fails to distinguish between a removed building, which represents a real change, and a building that simply appears darker due to a seasonal shift, leading to numerous false alarms [4], [20]. While asymmetric cross-interaction-based feature subtraction methods [14], [15], [18], [19], [20], [21] attempt to first align these features and then compute the feature difference, it introduces two critical bottlenecks: 1) *metric asymmetry*, where the divergence varies with transfer direction ($(t_1 \rightarrow t_2) \neq (t_2 \rightarrow t_1)$), violating the symmetry axiom of rigorous metrics; and 2) *artificial artifacts*, where forcing one image to strictly mimic the style of the other in pursuit of symmetry often generates fake details (e.g., artificial noise). These nonexistent patterns act as structural noise, making it difficult to precisely identify real object boundaries.

To overcome the theoretical bottlenecks in spatial representation and bitemporal change perception, we propose the phase-dominant structural sensing network (PDSS-Net). Diverging from spatial-only approaches, PDSS-Net addresses change detection (CD) in the frequency domain. Its core philosophy is disentanglement: it explicitly separates stable structural signals (encoded in the phase) from unstable seasonal styles (manifesting in the amplitude). Structurally, this unified framework comprises two coupled stages. First, to extract robust monotemporal representations, we propose the geometric-aware frequency alignment (GFA) module. Instead of treating the frequency spectrum as a rigid grid, GFA uses physics-aware log-polar coordinates (PLCs) to align receptive fields with the radial decay of frequency energy. Furthermore, by integrating anisotropic context decoupling (ACD) and phase-guided purification, it explicitly isolates directional structural ridges and protects intrinsic topological details from seasonal amplitude noise.

Second, for robust bitemporal change perception, we devise the bitemporal structure-guided frequency difference (BSFD) module. Its core symmetric consensus style learning (SCSL) mechanism realizes the symmetric spectral consensus highlighted in the title. SCSL symmetrizes the two temporal interaction directions and projects the bitemporal features into a shared, neutral amplitude space, thereby reducing temporal-order bias and amplitude-dominated intertemporal discrepancies. Crucially, to prevent the over-smoothing of genuine changes during this harmonization, we integrate a structure-guided protection scheme. By leveraging phase congruency to selectively boost structural energy, it ensures that high-frequency boundaries remain intact while seasonal discrepancies are neutralized.

The main contributions of this article are summarized as follows.

1) We propose the *PDSS-Net*, which reformulates CD as a frequency-domain disentanglement task, prioritizing stable phase consistency over unstable amplitude to resist phenological shifts.

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2) We develop the *GFA* module to rectify spectral geometric bias. By utilizing physics-aware coordinates and anisotropic filtering, it aligns receptive fields with radial energy distributions to precisely capture directional structural ridges.
3) We devise the *BSFD* module, whose core *SCSL* mechanism realizes the symmetric spectral consensus highlighted in the title. SCSL symmetrizes the two temporal interaction directions and maps both features to a shared consensus amplitude, while the structure-guided protection scheme safeguards genuine change boundaries during style harmonization.

The remainder of this article is organized as follows. Section II reviews the relevant literature on monotemporal semantic learning and bitemporal change perception. Section III provides a comprehensive description of the proposed PDSS-Net, including the GFA and BSFD modules. Section IV presents the experimental setup, quantitative results, and detailed ablation studies. Finally, Section V concludes this article.

II. Related Works

Current research paradigms in CD can be broadly categorized into two main streams: monotemporal semantic learning, which emphasizes extracting robust feature representations from individual images to mitigate environmental noise, and bitemporal change perception, which focuses on modeling intricate spatiotemporal dependencies to distinguish genuine changes from pseudo-changes. Accordingly, the proposed *PDSS-Net* is engineered to overcome the inherent limitations of these two critical stages, prompting a literature review from these two distinct perspectives.

A. Monotemporal Semantic Learning

To mitigate environmental interference in monotemporal features, existing strategies can be categorized into four classes based on their specific mechanisms.

1) *Semantic Reasoning and Context Consistency:* These methods mine deep semantics to resist local noise. Bi-SRNet [22] and SCDTN [23] incorporate reasoning blocks or late-fusion subnetworks to enforce internal semantic consistency. To balance detail and context, ELGC-Net [12] aggregates local–global features via pooled-transpose attention. Simultaneously, approaches like CTD-Transformer [7] and RHighNet [24] enhance the Siamese feature basis by mapping images to semantic tokens or applying spatial–channel attention. Others, such as DGMA2-Net [25], rely on the inherent robustness of deep backbones to filter radiometric differences.
2) *Object, Edge, and Structure Awareness:* These methods leverage physical structure invariance against style shifts. Recognizing that structural boundaries are resilient to seasonal changes, EGRCNN [6] and GSSR-Net [26] prioritize contour extraction and multiscale convolution. At the object level, CACG-Net [27] shifts focus from pixel differences to object shapes, while Vector-CNN [28] utilizes historical vector boundaries to constrain segmentation, facilitating robust object-oriented analysis.
3) *Large Model Adaptation and New Architectures:* Recent advancements exploit the generalization power of foundation models (e.g., SAM) and the new efficient global modeling of architectures (e.g., Mamba) to refine single-temporal representation. ChangeMamba [19] utilizes visual state-space models (VSSMs) for global dependencies, complemented by efficiency mechanisms in CEBlock [29]. In foundation model adaptation, FILFArch [30] aligns SAM encoders with remote sensing data via LoRA. Furthermore, Change-Agent [31] and CKCD [32] integrate linguistic semantics to deepen understanding beyond pure visual information.
4) *Specialized Feature Interaction:* Complementary strategies focus on calibrating features and refining input quality to mitigate environmental interference. BiFA [3] introduces a "mutual style calibration" mechanism, referencing cross-temporal information during encoding to actively suppress illumination differences, whereas CACD2Net [33] relies on the inherent high-level semantic invariance of deep Siamese networks to resist environmental discrepancies. At the data preprocessing level, SFBS-FFGNET [34] and CVA-AWGMM [35] prioritize signal quality; the former applies slow feature analysis to preselect invariant versus changing spectral bands, while the latter employs a neighborhood search strategy to correct geometric distortions and align physical positions. Distinct from Siamese structures, RDP-Net [36] utilizes a ConvMixer backbone within an early fusion framework, leveraging a region division layer to efficiently capture local detailed features.

Closely related to our frequency-domain motivation, CAIFNet [37] explicitly targets illumination-induced pseudo-changes by learning amplitude-invariant representations, exploiting the complementary roles of amplitude and phase to reduce the sensitivity of bitemporal features to appearance variations. This line of work demonstrates the effectiveness of frequency-domain disentanglement for robust CD. In contrast, PDSS-Net further treats the spectrum as a geometrically nonuniform field: it explicitly models radial and directional spectral structure through physics-aware coordinates and anisotropic filtering, and subsequently performs phase-guided purification rather than relying only on amplitude invariance.

Our method aligns with the fourth category but addresses three critical limitations in existing spatial and spectral approaches. First, unlike spatial methods that suffer from structure-style entanglement, we propose a frequency-domain disentanglement framework to explicitly separate phenological patterns from geometric structures. Second, to overcome the flaw where standard frequency methods ignore nonuniform spectral distributions, we introduce a physics-aware coordinate transformation that aligns with the inherent energy properties of remote sensing imagery. Finally, addressing the tradeoff between noise removal and detail preservation, we develop a phase-guided mechanism that filters seasonal interference

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without eroding faint structural edges, ensuring robustness against complex phenological shifts.

B. Bitemporal Change Perception

To effectively discriminate genuine changes from environmental variations, bitemporal analysis has evolved from elementary feature subtraction to sophisticated interaction mechanisms. We categorize these advancements into four distinct groups.

1) *Feature Interaction and Relation Modeling:* These methods mine spatiotemporal dependencies beyond arithmetic operations. Bi-SRNet [22] and TFI-GR [38] introduce reasoning blocks or interaction modules to capture semantic correlations and align feature distributions across time. To enhance nonlinearity, RHighNet [24] utilizes recursive gated convolutions for high-order feature extraction. In the frequency domain, FILFArch [30] leverages LoRA-based interaction to distinguish style from object changes. Furthermore, Change-Agent [31] and CKCD [32] integrate iterative refinement or cross-modal affinity, injecting textual guidance to activate key change regions.
2) *Sequence Modeling and Global Context:* Leveraging the sequential nature, these approaches utilize Transformers, SSMs, or RNNs. CTD-Transformer [7] and VcT [8] employ attention mechanisms to model object relationships or align background context. Recently, Mamba-based architectures like ChangeMamba [19] and GSSR-Net [26] apply spatiotemporal SSMs to learn evolution laws or establish long-range dependencies via omnidirectional scanning. Similarly, RNN-based models like EGRCNN [6] and STADE-CDNet [20] utilize LSTM or temporal memory modules to assess continuity.
3) *Guided Refinement and Alignment Strategies:* These methods utilize priors—such as flow fields or magnitude maps—for guidance. BiFA [3] explicitly estimates pixel offsets via differential flow fields for spatial warping. For boundary refinement, CACG-Net [27] and CACD2Net [33] leverage candidate objects or coarse maps to guide edge texture focus. In addition, approaches like DGMA2-Net [25] and Lv et al. [39] introduce multiscale fusion weights or precalculated magnitude maps as soft masks to highlight potential changes.
4) *Statistical Optimization and Enhancement:* Complementary approaches focus on loss functions and feature engineering. CEBlock [29] and SCDTN [23] optimize gradient contributions or triplet distances to balance samples. RDP-Net [36] introduces curriculum learning for detail preservation. In feature engineering, SFBS-FFGNET [34] and CVA-AWGMM [35] utilize difference matrices or adaptive Gaussian mixtures to enrich details. Finally, Vector-CNN [28] adopts postclassification comparison with historical vector rules.

Recent studies have also reconsidered change perception from the perspectives of human-inspired comparison and cross-task knowledge transfer. Li et al. [40] emulated the human difference-analysis process by constructing property-oriented descriptors and selectively comparing discriminative cues such as boundary and texture information, thereby suppressing misleading properties caused by illumination or appearance variations. In a broader semantic CD setting, Li et al. [41] further investigated transferring and adapting binary CD strategies to semantic CD, highlighting the value of reusable change-sensitive priors across related change interpretation tasks. These studies reinforce the importance of separating reliable structural evidence from unstable appearance cues when constructing bitemporal differences.

Our method aligns with the third category but addresses the following critical gaps in current alignment paradigms. First, existing strategies often rely on spatial warping or asymmetric style transfer, which introduces metric asymmetry and geometric distortion. We resolve this by employing a symmetric consensus projection that maps bitemporal states into a shared, neutral amplitude manifold, eliminating bias without distorting geometry. Second, while style alignment effectively removes seasonal differences, it risks hallucinating textures or over-smoothing genuine changes. To counteract this, we introduce a structure-guided protection mechanism that actively shields high-energy ridges during harmonization.

III. Method

A. Overview

Fig. 3 illustrates the overall pipeline of the proposed PDSS-Net. The bitemporal inputs I^{t_A} and I^{t_B} are processed by a weight-shared feature extractor Ψ_l, where the subscript l denotes the hierarchical feature level and indicates the specific operational unit applied at that level. In the following, we provide a structured overview of the framework, proceeding from left to right, consistent with the input-to-output flow as depicted. Initially, the bitemporal input images I^{t_A} and I^{t_B} are processed by a weight-sharing backbone, described as *hierarchical feature extraction* in Section III-B, to generate a set of basic multiscale hierarchical features $F_l^{t_k}$, where $l \in \{1, 2, 3, 4\}$ and $k \in \{A, B\}$.

Subsequently, these hierarchical feature sets are individually fed into the *GFA* detailed in Section III-C. It is worth noting that while this module shares weights across the temporal dimension to ensure consistency, it employs independent parameters for each scale level l to accommodate scale-specific characteristics. The primary objective of this design is to extract environment-robust representations from frequency spectra that are severely corrupted by seasonal and phenological interferences. Upon completion of this stage, we obtain the refined feature set $\{F_{l,out}^{t_k}\}$.

Following this, feature pairs from different timestamps but identical scales are forwarded to the *BSFD* in Section III-D. Operating on the mathematical principle of shared style (amplitude) and subtractive structure (phase), this module generates a set of change candidate features, denoted as $D_{l,CD}$.

Specifically, FD_1–FD_4 in Fig. 3 denote the same Feature Difference operation applied at four hierarchical feature levels, rather than four different feature types. Their distinction lies in the semantic scale of the corresponding inputs: shallow

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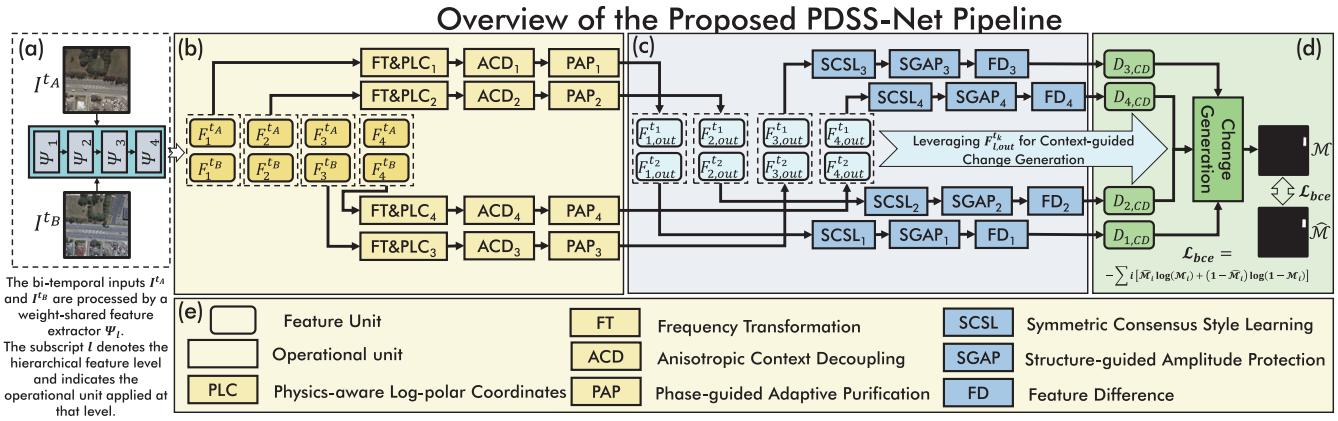


Fig. 3. Overview of the proposed PDSS-Net pipeline. The framework consists of four primary stages. (a) Hierarchical feature extraction for multiscale representation. (b) GFA to rectify spectral geometric bias. (c) BSFD to resolve interaction asymmetry. (d) Change generation based on hierarchical features for final mask reconstruction. (e) Legend of key operational units throughout the architecture.

levels retain finer spatial and boundary details, whereas deeper levels encode progressively more abstract semantic information. The resulting multiscale change features $D_{1,CD}$–$D_{4,CD}$ are subsequently integrated in the hierarchical Change Generation Stage, where complementary information across different scales is progressively aggregated for final prediction.

Finally, both the change candidates $D_{l,CD}$ and the refined features $\{F^{t_k}_{l,out}\}$ are integrated into the *change generation based on hierarchical features* in Section III-E to produce the final change map $\mathcal{M}$.

B. Hierarchical Feature Extraction

For a bitemporal image pair $\{I^{t_A}, I^{t_B}\} \in \mathbb{R}^{3\times H\times W}$, a ResNet18-based [9] backbone is employed to extract multilevel hierarchical features. The extraction process for each temporal image $I^{t_k}(k \in \{A, B\})$ is formulated as

$$\begin{aligned} F_0^{t_k} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{7\times 7}\left(I^{t_k}\right)\right)\right) \\ F_l^{t_k} &= \Psi_l\left(F_{l-1}^{t_k}\right), \quad l \in \{1,2,3,4\} \end{aligned} \tag{1}$$

where $F_0^{t_k}$ is the initial stem feature from $\mathrm{Conv}_{7\times7}(\cdot)$, Ψ_l denotes the lth residual stage and $\mathrm{BN}(\cdot)$ represent the batch normalization. The resulting hierarchical feature maps $F_l^{t_k} \in \mathbb{R}^{C_l\times H_l\times W_l}$ capture multiscale semantic information with progressive channel depth and cumulative strides.

C. Geometric-Aware Frequency Alignment

We propose the *GFA* to extract environment-robust representations from seasonally corrupted spectra. Treating the frequency domain as a nonstationary energy field, GFA synergizes *PLCs* and *ACD*. This design aligns the receptive field with the spectrum's radial decay and directional ridges. Complementing this, phase-guided adaptive purification (PAP) uses phase topology to modulate soft-thresholding, strictly preserving faint structural edges against seasonal noise. Fig. 4 illustrates the difference in feature semantic embeddings before and after GFA. Specifics are detailed below.

1) Frequency Transformation and PLCs: Traditional frequency-domain enhancements rely on static, uniform grids, failing to adapt to the distinct energy distributions of environmental noise and object boundaries in bitemporal imagery. To address this, we reconceptualize the Fourier domain as a nonstationary energy field. As visualized in Fig. 5, the amplitude spectra inherently exhibit a radial, nonuniform distribution centered at the origin. By explicitly mapping this geometric

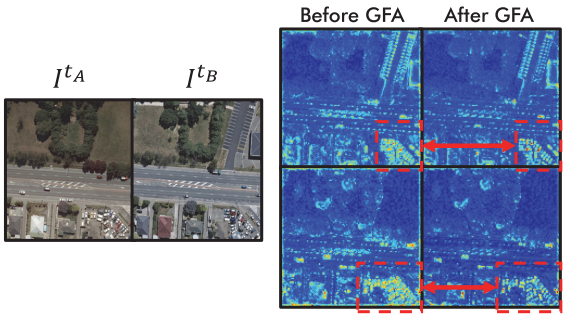


Fig. 4. GFA remarkably improves the semantic distinctiveness of building structures by isolating intrinsic topological details from environmental interference.

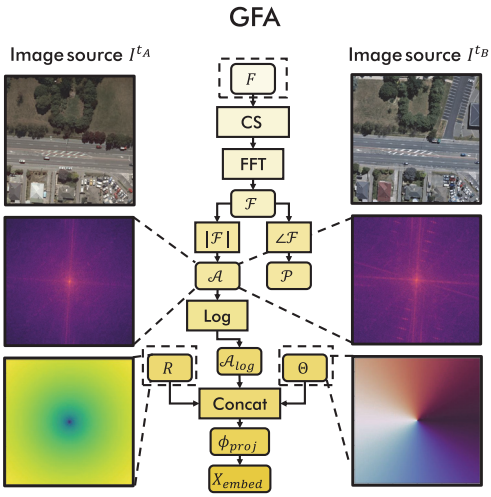


Fig. 5. Pipeline of the GFA module and visualization of 2-D amplitude and phase spectra. The frequency energy exhibits a radial distribution with orthogonal or star-shaped structural ridges; thus, applying conventional square kernels (e.g., 3 × 3) via translation is physically inconsistent with such a non-Euclidean domain. Note: primary inputs and outputs are highlighted with dashed boxes to clarify the workflow, a notation maintained across all subsequent modules.

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prior, our approach enables the precise disentanglement of phenological styles (dominant amplitude) from stable topological details (stable phase).

Furthermore, raw frequency representations are often ill-conditioned for CD due to the heavy-tailed $1/f$ energy distribution [42], [43], [44]. In bitemporal scenes, high-magnitude low-frequency components—representing seasonal backgrounds—tend to overshadow faint structural details. This energy imbalance hinders fine-grained gradient learning. To resolve this, we apply a logarithmic transformation to compress the dynamic range, enforcing feature equilibrium across the frequency spectrum before geometric projection.

Formally, let F be a generic feature tensor from the multi-scale temporal set $\{F_l^{t_k}\}$ [see (1)]. The following operations are applied independently to each scale. We first employ the 2-D fast Fourier transform (FFT(·)) to project F into the frequency domain, followed by a centering shift operation (CS(·)) to align the zero-frequency component with the spectrum center

$$\mathcal{F} = \mathrm{CS}\left(\mathrm{FFT}\left(F\right)\right). \tag{2}$$

Subsequently, we decompose the complex field $\mathcal{F}$ into its constituent amplitude spectrum $\mathcal{A}$ and phase spectrum $\mathcal{P}$. To rectify the heavy-tailed energy distribution (the $1/f$ law discussed prior), a logarithmic compression $\log(\cdot)$ is applied to the amplitude

$$\mathcal{A} = |\mathcal{F}|, \quad \mathcal{P} = \angle\mathcal{F}, \quad \mathcal{A}_{log} = \log\left(1 + \mathcal{A}\right) \tag{3}$$

where $|\cdot|$ and $\angle$ denote the modulus and argument operators, respectively. Here, $\mathcal{A}_{log}$ serves as the structurally balanced amplitude representation, effectively mitigating the dominance of low-frequency direct current components.

As discussed, standard convolutions are suboptimal for Fourier space due to their translation invariance. This weight-sharing mechanism imposes uniform filtering that conflicts with the location-dependent semantics of the nonstationary frequency field. To reconcile this mismatch, we introduce PLC to inject a geometric inductive bias. Acting as an explicit coordinate prior, PLC facilitates position-adaptive modulation, empowering the network to selectively attenuate phenological amplitude fluctuations in the low-frequency core while strictly safeguarding fine-grained structural textures in the high-frequency periphery.

Formally, we construct a coordinate anchor tensor $\mathcal{C}_{plc} = [R, \Theta]$ to encode the geometric properties of the frequency field (see Fig. 5 for the visualization of R and Θ). A logarithmic radial component R is employed to balance the sampling density across the radially decaying energy distribution, while an angular component Θ captures orientation-specific semantics

$$R\left(u, v\right) = \frac{\log\left(1 + \sqrt{u^2 + v^2}\right)}{\rho_{max}}, \quad \Theta\left(u, v\right) = \frac{\arctan\left(v/u\right)}{\pi} \tag{4}$$

where (u, v) denote the relative Cartesian coordinates defined with respect to the zero-frequency center, and ρ_{max} serves as the normalization constant representing the maximum Euclidean radius of the spectrum.

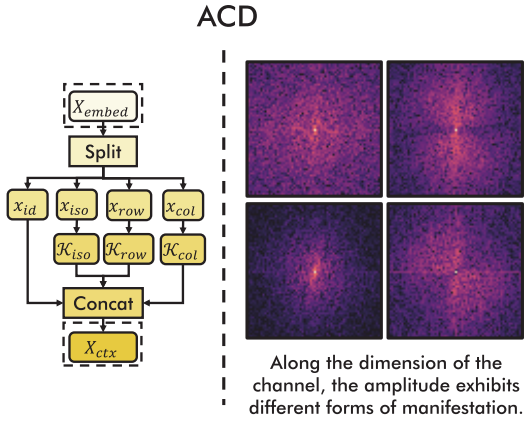


Fig. 6. Detailed structure of the ACD module and its spectral manifestations. (Right) Distinct amplitude preferences across different feature channels. These spectral representations are derived from the same bitemporal input example shown in Fig. 5.

To synthesize a unified spectral-geometric representation, we fuse the coordinate tensor $\mathcal{C}_{plc}$ with the log-compressed amplitude $\mathcal{A}_{log}$ via a linear projection operation

$$X_{embed} = \phi_{proj}\left(\mathrm{Concat}\left(\mathcal{A}_{log}, \mathcal{C}_{lpsc}\right)\right) \tag{5}$$

where Concat(·) denotes the channelwise concatenation. The operator ϕ_{proj} is a 1×1 convolutional fusion layer that projects the concatenated features back to the original channel dimension of $\mathcal{A}_{log}$, embedding the spatial coordinate priors into the frequency feature field.

2) Anisotropic Context Decoupling: To effectively decode the complex topology of the frequency field across different time nodes, we propose the ACD strategy. In bitemporal analysis, standard isotropic convolutions fail to distinguish stable structural ridges from transient seasonal noise. Moreover, as shown in Fig. 6, different frequency-domain channels exhibit markedly disparate spectral distribution preferences. To address this, rather than processing the bitemporal embeddings X_{embed} as monolithic entities, we stratify the feature space into four functional subspaces $\{x_{id}, x_{iso}, x_{row}, x_{col}\}$. This stratification acts as a frequency-selective sieve, isolating stable identity mappings (x_{id}) from directional components [9], [45]. It enables the network to adaptively rectify temporal misalignments by prioritizing orientation-specific structural features over omnidirectional environmental interference. We then route these decoupled components through a parallel sensing group

$$X_{ctx} = \mathrm{Concat}\left(x_{id}, \mathcal{K}_{iso}\left(x_{iso}\right), \mathcal{K}_{row}\left(x_{row}\right), \mathcal{K}_{col}\left(x_{col}\right)\right) \tag{6}$$

where the operators are designed to capture distinct energy topologies: $\mathcal{K}_{iso}$ (3×3 convolution) functions as an isotropic context aggregator to capture nondirectional local coherence; $\mathcal{K}_{row}$ (1×7) and $\mathcal{K}_{col}$ (7×1) serve as anisotropic resonance filters. These asymmetric operators align with the elongated frequency signatures of structural boundaries, effectively disentangling directional geometric priors from seasonal amplitude clutter. This mechanism serves as a frequency-domain manifestation of the disentanglement principle [46], [47], [48].

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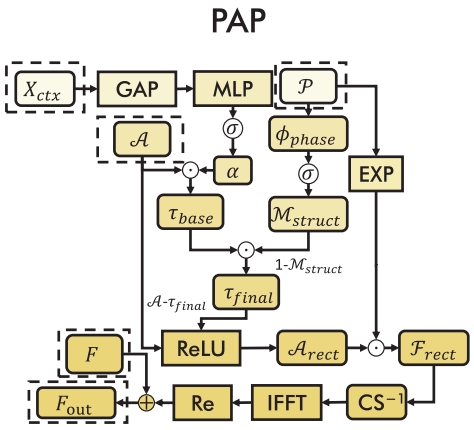


Fig. 7. Feature flow within the PAP module. It takes the anisotropic context (X_{ctx}), raw amplitude ($\mathcal{A}$), phase spectrum ($\mathcal{P}$), and original spatial feature (F) as inputs, and outputs the rectified feature (F_{out}).

3) Phase-Guided Adaptive Purification: Having extracted the anisotropic feature X_{ctx}, the subsequent objective is to purge the frequency representation of phenological contamination. Although *soft-thresholding* serves as a foundational paradigm for denoising [49], [50], [51], [52], its exclusive reliance on magnitude introduces a fundamental ambiguity: both transient seasonal noise (e.g., atmospheric scattering) and indispensable structural nuances (e.g., high-frequency edges) frequently manifest as low-amplitude components, leading to potential oversmoothing. Consequently, a uniform magnitude-based threshold risks inadvertently erasing the very structural topology we aim to preserve.

To resolve this dilemma, we leverage the phase spectrum as an auxiliary modality, capitalizing on its inherent ability to encode structural topology. Grounded in this physical axiom, our proposed PAP scheme (in Fig. 7) utilizes phase as a structural validator to differentiate between noise and genuine edges. This mechanism enables the selective suppression of low-energy noise while ensuring the preservation of subtle structural features that would otherwise be lost during amplitude harmonization. First, we aggregate the anisotropic frequency context X_{ctx} via global average pooling (GAP($\cdot$)) to predict channelwise scaling factors α, generating a baseline noise floor τ_{base}

$$\alpha = \sigma\left(\mathrm{MLP}\left(\mathrm{GAP}\left(X_{ctx}\right)\right)\right), \quad \tau_{base} = \alpha \odot \mathcal{A} \tag{7}$$

where σ represents the Sigmoid activation, and $\odot$ denotes the Hadamard product.

Applying the baseline threshold τ_{base} uniformly across the frequency-domain risks indiscriminately eroding faint details. To mitigate this, we extract a *structural integrity map* $\mathcal{M}_{struct}$ from the phase component $\mathcal{P}$. This design is grounded in the fundamental signal processing axiom [16], [17] that phase information strictly encodes the spatial location of image discontinuities (e.g., edges and textures), whereas amplitude primarily dictates intensity. Leveraging this topological stability as a robust reference, we employ a lightweight encoder ϕ_{phase} to project the phase prior into a gating weight

$$\mathcal{M}_{struct} = \sigma\left(\phi_{phase}\left(\mathcal{P}\right)\right) \in (0,1) \tag{8}$$

where $\mathcal{M}_{struct} \to 1$ indicates regions with high structural dominance.

We then modulate the noise threshold to be structure-aware

$$\tau_{final} = \tau_{base} \odot \left(1 - \mathcal{M}_{struct}\right). \tag{9}$$

This modulation ensures that in regions with strong phase-defined topology (where $\mathcal{M}_{struct} \approx 1$), the shrinkage threshold τ_{final} approaches zero, thereby strictly preserving the original amplitude energy. Conversely, in structure-agnostic regions, the full purification threshold is applied.

The rectified amplitude is obtained via the topology-protected soft-thresholding

$$\mathcal{A}_{rect} = \delta\left(\mathcal{A} - \tau_{final}\right) \tag{10}$$

where δ denotes the ReLU activation. This operation effectively forces background components with a low signal-to-noise ratio (SNR)—such as seasonal scattering—to zero, while strictly safeguarding significant structural peaks validated by the phase topology.

Then, to strictly preserve spatial structure, we reconstruct the complex frequency field $\mathcal{F}_{rect}$ by directly coupling the purified amplitude $\mathcal{A}_{rect}$ with the preserved original phase $\mathcal{P}$

$$\mathcal{F}_{rect} = \mathcal{A}_{rect} \odot \exp\left(j\mathcal{P}\right). \tag{11}$$

The final spatial feature map is restored via the inverse discrete Fourier transform (IFFT($\cdot$)) with a residual connection

$$F_{out} = F + \mathrm{Re}\left(\mathrm{IFFT}\left(\mathrm{CS}^{-1}\left(\mathcal{F}_{rect}\right)\right)\right) \tag{12}$$

where F_{out} denotes the final restored spatial feature maps, CS^{-1} represents the inverse centering operator, which realigns the frequency components to their canonical positions for the inverse discrete Fourier transform (IFFT($\cdot$)), and Re($\cdot$) extracts the real component to ensure the output remains in the real-valued spatial domain.

By applying this procedure to the multiscale features $F_l^{t_k}$ ($l \in \{1,2,3,4\}, k \in \{A,B\}$) from (1), we derive the corresponding refined feature set $\{F_{l,out}^{t_k}\}$.

D. Bitemporal Structure-Guided Frequency Difference

In bitemporal CD, phenological shifts manifest as global amplitude discrepancies rather than additive noise. Standard spatial subtraction fails to disentangle these multiplicative modulations from genuine structural phase shifts, inevitably conflating seasonal dimming with physical land-cover removal. To mitigate such pseudo-changes, recent approaches align temporal styles via directional cross interaction [14], [15], [18], [19], [20], [21] (e.g., $A \to B$). However, this paradigm introduces inherent metric asymmetry ($(A \to B) \neq (B \to A)$)—violating the symmetry axiom of rigorous distance metrics—and frequently induces texture hallucinations that distort true geometric boundaries.

To resolve this, we propose *SCSL*. As illustrated in Fig. 8, bitemporal inputs exhibit pronounced amplitude disparities across various frequency bands. Instead of forcing unidirectional alignment, SCSL addresses this by projecting both features into a shared, consensus amplitude space. However, such global harmonization risks inadvertently suppressing

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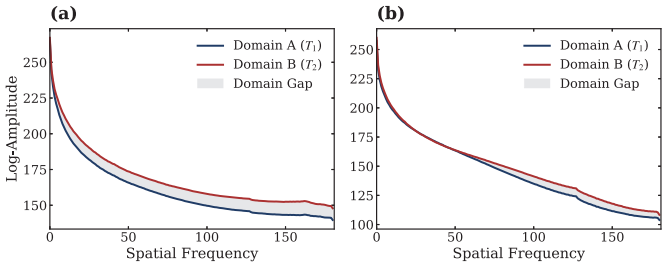


Fig. 8. Spectral amplitude distributions of (a) BCDD and (b) CLCD datasets. The curves illustrate the log-amplitude across the spatial frequency spectrum for dual-temporal images (T_1 and T_2). The shaded gray area quantifies the domain gap (spectral shift) between the two time points.

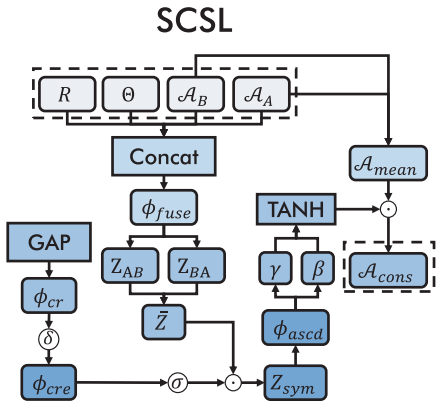


Fig. 9. Feature flow within the SCSL module. It takes the bitemporal log-amplitude spectra ($\mathcal{A}_A$ and $\mathcal{A}_B$) and coordinate anchors ($\{R, \Theta\}$) as inputs, and outputs the rectified consensus amplitude ($\mathcal{A}_{cons}$).

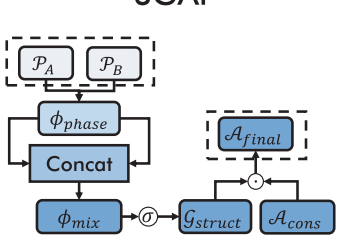


Fig. 10. Feature flow within the SGAP module. It takes the bitemporal phase spectra ($\mathcal{P}_A$ and $\mathcal{P}_B$) and consensus amplitude ($\mathcal{A}_{cons}$) as inputs, and outputs the final amplitude ($\mathcal{A}_{final}$).

high-frequency structural details. To counteract this side effect, we introduce structure-guided amplitude protection (SGAP). This mechanism actively preserves essential structural representations, preventing them from being over-smoothed during the feature calibration process.

Here, SCSL constitutes the core spectral-consensus operation of BSFD and corresponds to the "symmetric spectral consensus" highlighted in the title. It is designed to reduce the intertemporal spectral-style discrepancy within a bitemporal image pair, rather than to perform conventional source-to-target domain adaptation. Specifically, SCSL constructs the two ordered interactions Z_{AB} and Z_{BA} and symmetrizes them through $\bar{Z} = 0.5(Z_{AB} + Z_{BA})$ to predict a shared consensus amplitude. Consequently, the two temporal features are compared under the same amplitude reference while retaining their original phase information, thereby reducing temporal-order bias and amplitude-induced pseudo-changes. Figs. 9 and 10 illustrate the feature flows within the SCSL and SGAP modules, respectively.

1) Symmetric Consensus Style Learning: For brevity, we omit the scale index l of $F^{t_k}_{l,out}$ in (12) and refer to the generic bitemporal inputs as $F^{t_k}_{out}$ (where $k \in \{A, B\}$). Let $\mathcal{A}_A, \mathcal{A}_B$ and $\mathcal{P}_A, \mathcal{P}_B$ be the corresponding log-amplitude and phase spectra of $F^{t_A}_{out}$ and $F^{t_B}_{out}$, and let $\{R, \Theta\}$ be the physics-aware coordinate anchors defined in (4). To establish a temporally unbiased style representation, our adaptive symmetric consensus scheme dynamically recalibrates bidirectional spectral interactions into a unified context Z_{sym}

$$\begin{aligned} Z_{AB} &= \phi_{fuse}\left(\text{Concat}\left(\mathcal{A}_A, \mathcal{A}_B, R, \Theta\right)\right) \\ Z_{BA} &= \phi_{fuse}\left(\text{Concat}\left(\mathcal{A}_B, \mathcal{A}_A, R, \Theta\right)\right) \\ Z_{sym} &= \bar{Z} \odot \sigma\left(\phi_{cre}\left(\delta\left(\phi_{cr}\left(\text{GAP}\left(\bar{Z}\right)\right)\right)\right)\right) \end{aligned} \tag{13}$$

where ϕ_{fuse}, ϕ_{cr}, and ϕ_{cre} denote the 1×1 convolutions for embedding, channel reduction, and expansion, respectively. $\bar{Z} = 0.5(Z_{AB} + Z_{BA})$ is the mean symmetric context, and GAP$(\cdot)$ is global average pooling. δ and σ represent ReLU and Sigmoid activations for generating channel attention, applied via elementwise multiplication ($\odot$).

Subsequently, the context Z_{sym} is processed by the proposed ACD [denoted as Φ_{acd} in (6)], which captures direction-dependent spectral correlations as distinct energy ridges. We then predict the affine transformation parameters γ (scale) and β (shift) to model the consensus style

$$\gamma, \beta = \phi_{ascd}\left(\Phi_{acd}\left(\mathbf{Z}_{sym}\right)\right) \tag{14}$$

where ϕ_{ascd} denotes the regression head implemented via a 1×1 convolutional layer, designed to project the spectral context into the affine parameter space.

The baseline consensus amplitude $\mathcal{A}_{cons}$ is generated by modulating the mean spectral energy of the two time steps

$$\begin{aligned} \mathcal{A}_{mean} &= \frac{\mathcal{A}_A + \mathcal{A}_B}{2} \\ \mathcal{A}_{cons} &= \mathcal{A}_{mean} \odot (1 + \tanh(\gamma)) + \beta \end{aligned} \tag{15}$$

where $\mathcal{A}_{mean}$ serves as the unbiased spectral baseline. Regarding the affine parameters: β represents the additive spectral shift (modeling brightness bias), while γ controls the multiplicative contrast. Notably, we employ the hyperbolic tangent function $\tanh(\cdot)$ to constrain γ within $(-1, 1)$. This design formulates a residual scaling mechanism, where $(1 + \tanh(\gamma))$ modulates the amplitude intensity within a stable range of $(0, 2)$, preventing numerical instability during consensus learning.

2) Structure-Guided Amplitude Protection: Distinct from the subtractive thresholding in PAP (see Section III-C3), this stage focuses on preserving structural details during bitemporal style harmonization. While $\mathcal{A}_{cons}$ effectively mitigates seasonal discrepancies, its unconstrained application risks suppressing genuine structural changes [15], [18], [21], [53], which similarly exhibit high-magnitude frequency energy.

To address this, we introduce the SGAP mechanism, leveraging the axiom that "phase congruency implies structural saliency." We employ a lightweight phase encoder ϕ_{phase} to

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project the raw phase information into a latent structural space. A structural consistency gate $\mathcal{G}_{struct}$ is then computed to evaluate local topological stability

$$\mathcal{G}_{struct} = \sigma\left(\phi_{mix}\left(\text{Concat}\left(\phi_{phase}\left(\mathcal{P}_A\right), \phi_{phase}\left(\mathcal{P}_B\right)\right)\right)\right) \quad (16)$$

where σ is the Sigmoid function and ϕ_{phase} denotes a 1×1 convolutional layer. A value of $\mathcal{G}_{struct} \to 1$ indicates regions where phase information strongly suggests a coherent physical structure.

We then modulate the consensus amplitude using this gate to perform selective energy boosting

$$\mathcal{A}_{final} = \mathcal{A}_{cons} \odot (1 + \lambda \mathcal{G}_{struct}) \quad (17)$$

where λ is a scaling factor (set to 0.5 in implementation; analysis experiments can be seen in Section IV-G2). Physically, this mechanism establishes a phase-dependent protection barrier. In regions dominated by incoherent seasonal noise where phase distributions are random, $\mathcal{G}_{struct}$ approaches zero, thereby enforcing the unified consensus style to suppress intertemporal spectral-style discrepancies. Conversely, in regions exhibiting strong structural edges characterized by coherent phase, $\mathcal{G}_{struct}$ approaches one, actively shielding the high-energy topological features from being over-smoothed by the harmonization process.

3) Feature Difference: With the feature representation now aligned in style and explicitly reinforced in structural integrity, we establish a feature difference mechanism to highlight change regions. This approach aligns with a prevalent paradigm in recent literature [1], [2], where the explicit modeling of bitemporal disparities serves as the cornerstone for robust CD.

Specifically, we project the bitemporal features back into the spatial domain by coupling the shared consensus amplitude $\mathcal{A}_{final}$ with their distinct original phase topologies $\mathcal{P}_A$ and $\mathcal{P}_B$

$$\begin{aligned} Y_A &= \text{IFFT}\left(\text{CS}^{-1}\left(\mathcal{A}_{final} \cdot e^{j\mathcal{P}_A}\right)\right) \\ Y_B &= \text{IFFT}\left(\text{CS}^{-1}\left(\mathcal{A}_{final} \cdot e^{j\mathcal{P}_B}\right)\right) \end{aligned} \quad (18)$$

where CS^{-1} denotes the inverse centering operator, and IFFT$(\cdot)$ represents the inverse discrete Fourier transform.

The final change intensity map is derived as the symmetric difference of these sensed structures

$$D_{CD} = |Y_A - Y_B|. \quad (19)$$

Mathematically, this shared-amplitude reconstruction guarantees style invariance. For seasonal pseudo-changes, the invariant structural topology ($\mathcal{P}_A \approx \mathcal{P}_B$) combined with the shared $\mathcal{A}_{final}$ yields identical reconstructions, driving the response to zero ($D_{CD} \to 0$). Conversely, genuine changes disrupt phase coherence ($\mathcal{P}_A \neq \mathcal{P}_B$), which are strictly preserved in D_{CD}. As shown in Fig. 11, this mechanism effectively suppresses confusing pseudo-changes (e.g., unchanged buildings) and highlights genuine structural evolution.

E. Change Generation Based on Hierarchical Features

To generate the final change map, we employ a hierarchical decoding architecture that progressively integrates the refined bitemporal features $\{F^{t_k}_{l,out}\}$ with the change-aware embeddings $D_{l,CD}$.

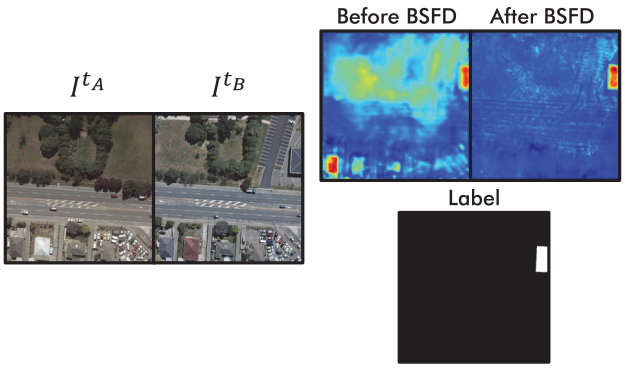


Fig. 11. Visualization of change embeddings before and after the BSFD module. The proposed mechanism effectively suppresses pseudo-changes and strictly aligns semantic perception with genuine structural evolution.

For each scale $l \in \{1, \ldots, 4\}$, the refined bitemporal features are first fused with the corresponding change-aware embedding through a shared fusion operator ϕ_l. To maintain information integrity, we employ a residual summation to derive the enhanced consensus features $\hat{D}_l$

$$\hat{D}_l = D_{l,CD} + \sum_{k \in A,B} \phi_l\left(\left[F^{t_k}_{l,out} \parallel D_{l,CD}\right]\right) \quad (20)$$

where $\parallel$ denotes the channelwise concatenation, and ϕ_l represents a 3×3 convolutional layer that projects the concatenated features back to the original channel dimension.

The enhanced features are then aggregated through a top-down cascaded pathway. Let $\mathcal{H}_l(\cdot)$ be the sequential convolution blocks and $\uparrow$ represent the bilinear upsampling operator. The reconstruction process is formulated as

$$Z_l = \begin{cases} \mathcal{H}_l\left(\hat{D}_l\right), & \text{if } l = 1 \\ \mathcal{H}_l\left(\hat{D}_l + (Z_{l-1} \uparrow)\right), & \text{if } l > 1. \end{cases} \quad (21)$$

This progressive integration ensures the stable propagation of high-level semantic information to low-level structural details. Finally, the change mask $\mathcal{M}$ is produced by projecting the most refined feature map Z_4 into the spatial probability domain

$$\mathcal{M} = \phi_{final}\left(Z_4\right) \quad (22)$$

where ϕ_{final} represents a 1×1 convolutional layer.

The network is optimized end-to-end using the standard binary cross-entropy loss to minimize the discrepancy between the prediction $\mathcal{M}$ and the ground truth $\hat{\mathcal{M}}$. Formally, the loss is defined as

$$\mathcal{L}_{bce} = -\sum i\left[\hat{\mathcal{M}}_i \log\left(\mathcal{M}_i\right) + \left(1 - \hat{\mathcal{M}}_i\right) \log\left(1 - \mathcal{M}_i\right)\right] \quad (23)$$

where i indexes all pixels of the input image.

IV. Experimental Results and Discussion

A. Experimental Procedure

To comprehensively evaluate the performance, robustness, and internal mechanisms of the proposed PDSS-Net, this section presents extensive experiments and detailed analyses.

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The remainder of this section is organized as follows. First, we detail the experimental datasets, implementation protocols, and evaluation metrics in Section IV-B. Subsequently, Sections IV-C and IV-D present thorough quantitative and qualitative comparisons against 18 state-of-the-art methods to demonstrate the overall superiority of our framework. To validate the rationality of our architectural design, Sections IV-E and IV-G conduct rigorous ablation studies and parameter sensitivity analyses on the core modules. Furthermore, Section IV-H provides an explicit physical verification of the model's structural fidelity in the frequency domain, while Section IV-I assesses its resilience against nonstationary environmental noise. Finally, Section IV-M evaluates the deployment potential by analyzing the tradeoff between computational efficiency and detection accuracy.

B. Setting

Datasets: Comprehensive experiments are conducted on six standard benchmarks: LEVIR-CD [58], GoogleBuilding [59], CLCD [54], DSIFN [60], SECOND [4], and BCDD [61]. Following a 3:1:1 split ratio, the training/validation/testing samples are partitioned as 1160/387/387, 500/167/167, 360/120/120, 2390/797/797, 1600/533/533, and 960/320/320, respectively. All images are uniformly resized to a spatial resolution of 512 $\times$ 512.

Implementation Details: The proposed framework was implemented in PyTorch and trained on an NVIDIA RTX 4090 GPU. We employed the Adam optimizer with an initial learning rate of 1×10^{-4} and a batch size of 4 for 200 epochs.

Architectural Hyperparameters for PDSS-Net: We configure a four-level hierarchy ($l \in \{1, 2, 3, 4\}$) based on (1) with the dimensions $C_l \times H_l \times W_l$ set to $32 \times 128 \times 128$, $64 \times 64 \times 64$, $128 \times 32 \times 32$, and $256 \times 16 \times 16$, respectively. The scaling factor λ in (17) is fixed at 0.5. For the ACD module in (6), the convolution kernel sizes $\mathcal{K}_{iso}$, $\mathcal{K}_{row}$, and $\mathcal{K}_{col}$ are defined as 3×3, 1×7, and 7×1.

Metric: Based on the four elements of the confusion matrix—true positive (TP), true negative (TN), FP (i.e., false alarms), and false negative (FN, i.e., missed detections)—we employ the following six metrics to quantitatively evaluate the detection performance.

Intersection Over Union (IoU): It strictly measures the spatial overlap between the predicted changes and the ground truth, penalizing both false alarms and missed detections simultaneously

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{24}$$

Precision: It reflects the reliability of the model's predictions and its ability to suppress false alarms (FP)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{25}$$

Recall: It reflects the model's ability to identify true change regions and resist missed detections FN

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{26}$$

$F1$-Score: The harmonic mean of precision and recall. Given that unchanged background pixels typically dominate remote sensing images (indicating severe class imbalance), the $F1$-score serves as the primary metric for evaluating the comprehensive detection performance on change targets

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{27}$$

Overall Accuracy (OA): It measures the global pixelwise classification accuracy. However, in CD tasks, this metric is less sensitive to actual changes due to the dominance of massive background regions (TN)

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{28}$$

FP Rate (FPR): It measures the proportion of unchanged pixels that are incorrectly predicted as changes, and therefore directly quantifies the pseudo-change false-alarm rate. A lower FPR indicates stronger pseudo-change suppression

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{29}$$

Fairness of Comparison Protocol: To ensure a strictly fair comparison, all 18 baseline methods reported in Tables I and II were not directly copied from their original papers. Instead, they were reproduced using their official open-source implementations and retrained under the exact same experimental protocol as our PDSS-Net. Specifically, all models were trained using the same dataset splits, identical data augmentation strategies, and optimized for the same total epochs. Furthermore, the input images for all comparative models were uniformly resized to a spatial resolution of 512 $\times$ 512. This specific resolution was chosen because it strikes an optimal balance between preserving sufficient high-frequency structural details and accommodating the computational memory constraints of our hardware during the FFT operations.

C. Quantitative Analysis

As presented in Tables I and II, we benchmarked the proposed method against 18 state-of-the-art CD approaches developed between 2018 and 2026. In addition to the commonly used IoU, $F1$-score, precision, recall, and OA, we further report the FPR to specifically quantify the suppression of pseudo-changes. FPR measures the proportion of unchanged pixels that are incorrectly predicted as changed; therefore, a lower FPR indicates fewer false alarms caused by background fluctuations, seasonal variations, illumination differences, or other nonsemantic disturbances. The experimental results demonstrate that PDSS-Net achieves highly competitive overall performance across all six datasets while consistently maintaining notably low FPR values, confirming its effectiveness in suppressing pseudo-changes without compromising the detection of genuine changes.

On the LEVIR-CD dataset, our method achieves state-of-the-art results across key metrics, securing an IoU of 84.61%, an $F1$-score of 91.66%, and an OA of 98.87%. It also obtains the lowest FPR of 0.46%, reducing the FPR by 0.19 percentage points compared with the second-best result of 0.65%,

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TABLE I

QUANTITATIVE COMPARISON RESULTS ON LEVIR-CD, CLCD, AND BCDD DATASETS. ALL VALUES ARE IN PERCENTAGE (%). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Model	LEVIR-CD						CLCD						BCDD					
	IoU	F1	Precision	Recall	FPR↓	OA	IoU	F1	Precision	Recall	FPR↓	OA	IoU	F1	Precision	Recall	FPR↓	OA
FCEF [5]	70.52	82.71	73.44	**94.66**	2.54	97.27	38.08	55.16	43.32	75.89	6.77	92.12	71.55	83.42	81.92	84.97	2.11	96.58
FCSiam [5]	69.25	81.83	72.31	94.24	2.68	97.11	37.60	54.65	44.84	69.96	5.87	92.59	76.14	86.45	82.56	90.73	2.16	97.12
MSCANet [54]	82.29	90.28	89.06	91.54	0.83	98.64	55.49	71.37	71.32	71.43	1.96	96.34	82.56	90.45	96.69	84.96	0.33	98.18
DGMAANet [25]	83.05	90.74	90.24	91.25	0.73	98.71	55.60	71.47	78.21	65.79	1.25	96.65	88.07	93.66	95.78	91.63	0.46	98.74
DAMFANet [55]	81.96	90.09	91.11	89.08	0.65	98.64	52.94	69.23	67.00	71.61	2.41	95.94	78.48	87.94	93.58	82.95	0.64	97.70
ECTI [56]	82.57	90.45	89.37	91.56	0.81	98.66	54.25	70.34	67.94	72.92	2.35	96.07	87.89	93.55	94.61	92.52	0.59	98.71
Diformer [13]	82.59	90.46	88.97	92.01	0.85	98.66	56.01	71.80	69.32	74.47	2.25	96.27	86.33	92.66	95.05	90.39	0.53	98.55
BIFA [3]	82.65	90.50	90.10	90.91	0.74	98.68	56.58	72.27	70.72	73.89	2.09	96.38	86.34	92.67	95.38	90.11	0.49	98.56
STADECDNet [20]	80.12	88.96	84.77	93.59	1.25	98.40	42.62	59.77	46.15	**84.78**	6.75	92.71	88.74	94.03	92.61	95.50	0.86	98.77
ELGCNet [12]	78.65	88.05	86.35	89.82	1.05	98.32	41.93	59.09	56.22	62.26	3.31	94.50	86.28	92.63	91.90	93.38	0.93	98.50
Changemamba [19]	80.81	89.39	89.29	89.48	0.80	98.53	57.74	73.21	78.94	68.25	1.24	96.81	82.18	90.22	93.40	87.25	0.69	98.08
DSFDCD [11]	81.88	90.04	88.12	92.04	0.92	98.59	36.94	53.95	44.56	68.36	5.80	92.55	71.05	83.08	82.51	83.65	2.00	96.55
RHighNet [24]	80.29	89.07	86.29	92.03	1.09	98.44	52.83	69.14	70.05	68.24	1.99	96.11	84.71	91.72	91.84	91.60	0.92	98.33
STRobustNet [53]	79.53	88.60	86.24	91.09	1.08	98.38	47.51	64.42	62.86	66.05	2.66	95.34	81.41	89.75	92.16	87.47	0.84	97.98
ISDANet [57]	80.75	89.35	86.06	92.90	1.12	98.47	49.60	66.31	69.47	63.43	1.90	95.89	83.42	90.96	91.42	90.51	0.96	98.18
CD-STMamba [18]	82.62	90.48	89.50	91.49	0.80	98.67	50.93	67.49	72.88	62.84	1.59	96.13	**91.38**	**95.50**	94.36	**96.66**	0.65	99.08
RACDNet [14]	82.47	90.39	89.22	91.60	0.82	98.65	46.70	63.67	58.01	70.55	3.48	94.86	87.73	93.46	94.35	92.59	0.63	98.69
WHFCE [15]	84.08	91.35	89.39	93.40	0.82	98.78	44.82	61.90	59.42	64.59	3.01	94.92	71.94	83.68	90.83	77.57	0.88	96.93
Ours (PDSS-Net)	**84.61**	**91.66**	**93.50**	89.90	**0.46**	**98.87**	**59.80**	**74.84**	**83.87**	67.57	**0.89**	**97.10**	91.22	95.41	**98.73**	92.30	**0.13**	**99.10**

TABLE II

QUANTITATIVE COMPARISON RESULTS ON GOOGLE, DSIFN, AND SECOND DATASETS. ALL VALUES ARE IN PERCENTAGE (%). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Model	Google						DSIFN						SECOND					
	IoU	F1	Precision	Recall	FPR↓	OA	IoU	F1	Precision	Recall	FPR↓	OA	IoU	F1	Precision	Recall	FPR↓	OA
FCEF [5]	66.98	80.23	79.61	80.85	4.01	93.53	62.11	76.63	69.18	85.87	14.18	85.84	33.02	49.65	45.48	54.65	15.89	78.36
FCSiam [5]	67.67	80.72	74.25	88.42	5.94	93.15	53.78	69.94	61.44	81.18	18.88	81.14	32.83	49.43	42.08	59.90	20.00	76.07
MSCANet [54]	85.99	92.47	91.71	93.24	1.63	97.54	89.24	94.31	95.72	92.95	1.54	96.97	48.75	65.55	63.89	67.29	9.23	86.19
DGMAANet [25]	87.21	93.17	94.39	91.98	1.06	97.81	89.34	94.37	95.28	93.48	1.72	96.98	51.67	68.13	69.81	66.54	6.98	87.85
DAMFANet [55]	85.18	92.00	93.15	90.87	1.29	97.43	89.06	94.21	95.90	92.59	1.47	96.93	48.78	65.57	61.47	70.26	10.68	85.60
ECTI [56]	87.68	93.44	92.19	94.72	1.55	97.84	91.02	95.30	95.32	95.28	1.73	97.46	49.52	66.24	62.72	70.18	10.12	86.03
Diformer [13]	86.35	92.68	91.67	93.70	1.65	97.60	91.26	95.43	95.57	95.29	1.64	97.53	49.38	66.11	63.59	68.85	9.56	86.22
BIFA [3]	85.83	92.37	93.53	91.25	1.22	97.56	88.71	94.02	**96.37**	91.78	**1.28**	96.84	48.56	65.37	60.77	70.73	11.08	85.37
STADECDNet [20]	81.89	90.04	83.36	**97.89**	3.78	96.49	71.05	83.08	75.17	92.84	11.36	89.77	49.07	65.83	57.61	**76.80**	13.71	84.44
ELGCNet [12]	74.40	85.32	81.67	89.31	3.88	95.01	88.74	94.03	94.82	93.26	1.89	96.80	50.41	67.03	66.84	67.22	8.09	87.09
Changemamba [19]	80.81	89.39	91.44	87.42	1.59	96.63	73.59	84.79	84.07	85.51	6.00	91.70	49.75	66.44	67.32	65.59	7.72	87.07
DSFDCD [11]	70.31	82.57	82.07	83.07	3.52	94.31	70.78	82.89	89.65	77.08	3.30	91.40	48.09	64.95	62.41	67.70	9.89	85.73
RHighNet [24]	77.98	87.63	83.43	92.27	3.55	95.77	92.42	96.06	96.02	96.10	1.48	**97.87**	44.95	62.02	60.50	63.62	10.08	84.79
STRobustNet [53]	80.49	89.19	89.93	88.46	1.92	96.52	83.05	90.74	95.55	86.39	1.49	95.23	45.27	62.33	58.72	66.40	11.33	84.33
ISDANet [57]	80.39	89.13	86.00	92.49	2.92	96.34	73.08	84.45	94.61	76.25	1.61	92.41	49.98	66.65	64.71	68.71	9.09	86.57
CD-STMamba [18]	84.58	91.65	90.58	92.74	1.87	97.26	86.06	92.51	93.60	91.44	2.32	96.00	49.13	65.89	66.35	65.43	8.05	86.77
RACDNet [14]	85.61	92.25	91.70	92.80	1.63	97.47	82.22	90.24	90.61	89.88	3.45	94.75	51.33	67.84	66.10	69.67	8.67	87.10
WHFCE [15]	80.64	89.28	89.10	89.47	2.12	96.51	74.36	85.29	87.79	82.94	4.27	92.27	48.34	65.17	58.90	72.95	12.35	84.78
Ours (PDSS-Net)	**90.68**	**95.11**	**98.05**	92.35	**0.36**	**98.46**	**92.47**	**96.09**	95.00	**97.20**	1.90	97.86	**53.07**	**69.34**	**81.58**	60.29	**3.30**	**89.59**

which directly demonstrates stronger suppression of pseudo-changes. When compared with the latest WHFCE model from 2026, we observe performance gains of 0.53% in IoU and a substantial 4.11% in Precision. This indicates that while maintaining a high detection rate, our approach is significantly more effective at suppressing pseudo-changes and reducing false alarms. In contrast to early CNN-based models such as FCEF and FCSiam, which exhibit extremely high recall rates exceeding 94% but suffer from low precision around 70%, our analysis suggests these models tend to trade precision for recall by over-identifying changes, resulting in considerable noise. Conversely, our method achieves an optimal balance, delivering a precision of 93.50% alongside a recall of 89.90%.

The CLCD dataset is widely considered challenging due to its complex farmland scenarios and fragmented change regions, evidenced by the fact that most models yield an IoU below 55%. Our method breaks this ceiling, achieving an IoU of 59.80% and an $F1$-score of 74.84%, which outperform the second-best model—Changemamba—by margins of 2.06% and 1.63%, respectively. Meanwhile, PDSS-Net achieves the lowest FPR of 0.89%, outperforming the second-best value of 1.24% and confirming its effectiveness in filtering farmland appearance variations that would otherwise be misclassified as changes. While many models that perform adequately on other datasets, such as DSFDCD and STADECDNet, suffer severe performance degradation on CLCD with IoU scores dropping below 45%, our method maintains robust performance. This stability is primarily attributed to our proposed frequency-domain geometric perception mechanism, specifically the PLCs. This mechanism effectively distinguishes between complex textural backgrounds, such as crop growth, and genuine structural changes, thereby demonstrating strong generalization capabilities in complex scenes.

Nevertheless, the IoU of 59.80% on CLCD remains lower than those on most of the other datasets, indicating that complex agricultural scenes remain challenging for PDSS-Net. CLCD [54] is a fine-grained cropland CD dataset, where crop growth, harvesting, and seasonal phenological variations often exhibit appearances similar to genuine land-cover changes. Meanwhile, variations in image quality and spatial resolution between bitemporal images further increase the difficulty of recognizing small changed regions and fine field boundaries. Although PDSS-Net achieves the best comparative performance on CLCD, incorporating stronger phenology-aware

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modeling and fine-boundary perception may further improve its robustness in complex agricultural scenes.

The BCDD dataset focuses on large-scale building changes with distinct boundaries. On this benchmark, our method achieves remarkable results, reaching a precision of 98.73% and an OA of 99.10%, both of which rank first among all compared methods. More importantly, its FPR is only 0.13%, substantially lower than the second-best value of 0.33%, indicating that the high precision is accompanied by a genuine reduction in pseudo-change false alarms. Notably, CD-STMamba achieves the highest recall of 96.66% and a marginal advantage in IoU with 91.38% compared with our 91.22%. This discrepancy reflects the distinct preferences of different architectures: Mamba-based models excel at capturing long-range dependencies, tending to fill building interiors more completely, which boosts recall. In contrast, our method prioritizes structural fidelity in the frequency domain, excelling in edge segmentation accuracy and noise resistance as evidenced by our superior precision. Although our IoU is lower by a slight margin of 0.16%, our OA is higher, indicating fewer total prediction errors.

On the Google dataset, characterized by high-resolution imagery and complex urban structures, our method demonstrates a commanding lead with state-of-the-art IoU and $F1$-scores of 90.68% and 95.11%, respectively. PDSS-Net further records the lowest FPR of 0.36%, compared with 1.06% for the runner-up, showing that it effectively rejects urban background variations and avoids spurious building-change responses. This performance surpasses the second-best ECTI model by significant margins of 3.00% in IoU and 1.67% in $F1$. Notably, our model attains an exceptional precision of 98.05%, outperforming the nearest competitor DGMAANet by nearly 3.66%. While STADECDNet achieves a higher recall of 97.89%, its significantly lower precision of 83.36% indicates a tendency toward over-segmentation. In contrast, our method maintains a robust recall of 92.35% while maximizing precision, ensuring highly reliable CD critical for urban monitoring.

On the DSIFN dataset, our model secures top-tier results, achieving the highest IoU of 92.47%, Recall of 97.20%, and the second-highest OA of 97.86%. Its FPR of 1.90% is also highly competitive and is only 0.62 percentage points above the best value of 1.28%, demonstrating that the improved Recall does not lead to a substantial increase in pseudo-change false alarms. While RHighNet posts a marginally lower $F1$-score, our method surpasses it in IoU and demonstrates a superior recall compared with the 96.10% achieved by RHighNet, indicating a stronger capability to capture complete change regions without missing subtle details. Furthermore, the competitive performance against specialized models like BIFA highlights our adaptability in effectively balancing precision and recall to deliver accurate segmentation maps.

Renowned for semantic ambiguity in complex urban environments, the SECOND dataset serves as a demanding benchmark where our method establishes a new state-of-the-art. We achieve an IoU of 53.07% and an $F1$-score of 69.34%, significantly outperforming the runner-up DGMAANet by margins of 1.40% and 1.21%. The FPR is reduced to 3.30%, far below the second-best result of 6.98%, providing direct quantitative evidence that PDSS-Net suppresses pseudo-changes caused by semantic confusion and background fluctuations. A striking observation is the precision gap, where our score of 81.58% vastly exceeds the 69.81% achieved by DGMAANet, underscoring exceptional discrimination between true semantic changes and background fluctuations. While STADECDNet records a higher recall of 76.80%, its precision drops drastically to 57.61%, reinforcing the common struggle to filter noise. Our superior OA of 89.59% confirms the effectiveness of our approach in handling these semantic complexities with high fidelity.

D. Visualization Analysis

To provide a comprehensive performance assessment, we calculated the mean IoU (mIoU) across all six datasets. Our method achieves a leading average mIoU of 78.64%, followed by the top-10 other methods, including DGMAANet at 75.82%, ECTI at 75.49%, Diformer at 75.32%, BIFA at 74.78%, CD-STMamba at 74.12%, MSCANet at 74.05%, DAMFANet at 72.73%, RACDNet at 72.68%, RHighNet at 72.20%, and ChangeMamba at 70.81%.

Figs. 12 and 13 visualize the above methods' comparative results, where red regions denote FPs and green regions indicate FNs. Specifically, on the CLCD dataset, most methods struggle due to the high semantic ambiguity between farmland changes and the background. In contrast, our method demonstrates remarkable robustness, with errors primarily confined to the subtle boundaries of disappearing or emerging farmland. On the Google dataset, while our approach exhibits some boundary delineation errors (green), it significantly minimizes pseudo-CD (red) compared with competing methods. This reduction in FPs underscores our model's superior semantic understanding and its ability to precisely target relevant change semantics. Similar observations are consistent across the remaining datasets.

E. Ablation Analysis

To validate the effectiveness of the proposed PDSS-Net components, we conducted extensive ablation studies across six datasets. As detailed in Table III, we evaluate the contributions of PLCs (M1), ACD (M2), PAP (M3), SCSL (M4), and SGAP (M5).

Effectiveness of the GFA (M1–M3): The GFA extracts environment-robust representations by rectifying spectral bias. *PLC (M1):* Removing PLC drops the IoU on the complex CLCD dataset from 59.80% to 58.14%. This confirms that adapting receptive fields to radial energy distributions effectively corrects geometric bias in heterogeneous textures. *ACD (M2):* Omitting ACD decreases IoU by 0.77% on LEVIR-CD and 1.59% on CLCD, demonstrating the necessity of anisotropic filtering to isolate directional structural ridges from omnidirectional noise. *PAP (M3):* Excluding PAP causes a severe 2.72% IoU drop on CLCD. This proves that relying solely on amplitude thresholding erases subtle edges, whereas phase-guided purification successfully preserves genuine structural topology. *GFA Removal:* To isolate the specific

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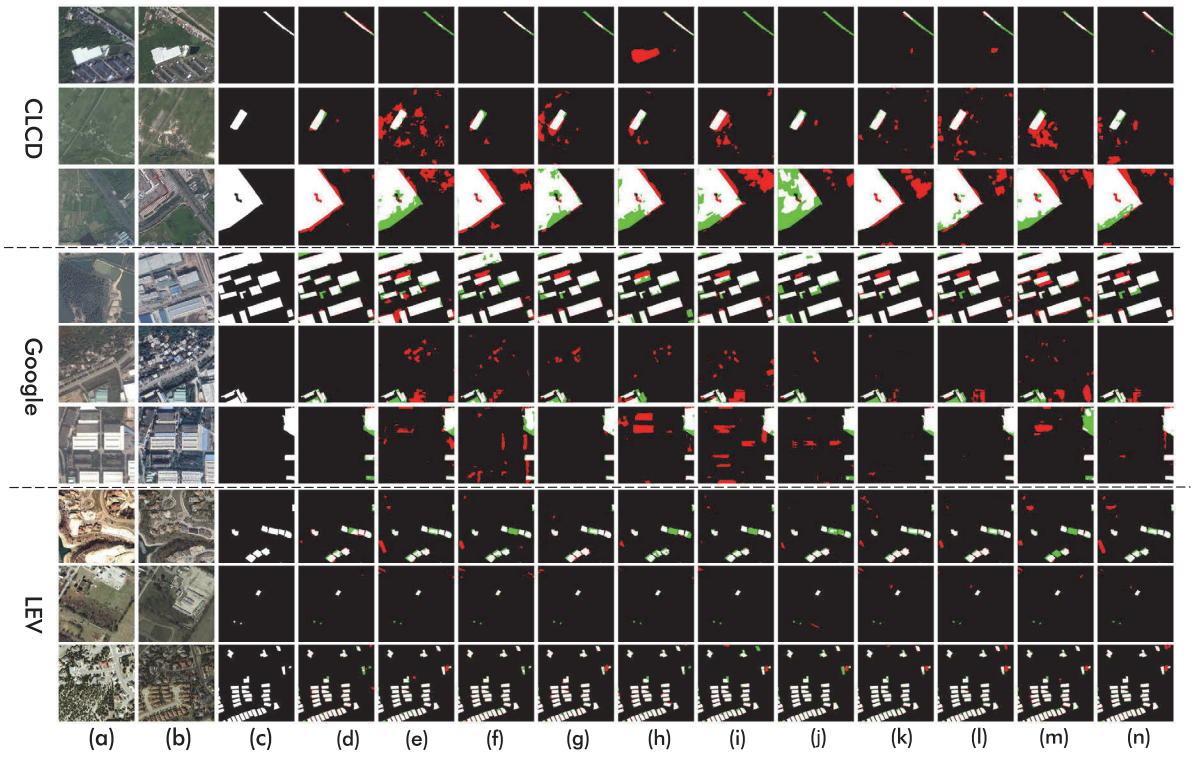


Fig. 12. (a)–(n) Input image at t_A, input image at t_B, ground truth, Ours, DGMAANet, ECTI, Diformer, BIFA, CD-STMamba, MSCANet, DAMFANet, RACDNet, RHighNet, and ChangeMamba. In the prediction maps, white denotes the TPs, black denotes the TNs, red denotes the FPs, and green denotes the FNs. The same color scheme is used in the following qualitative visualizations unless otherwise specified.

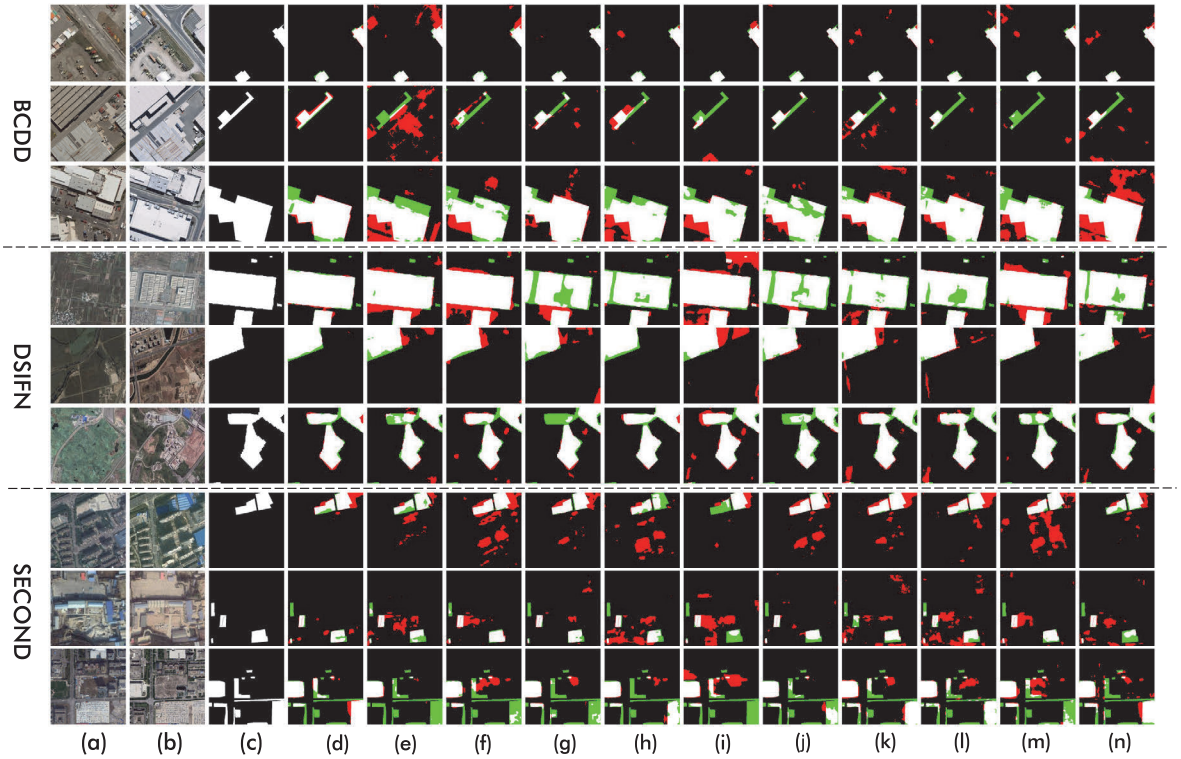


Fig. 13. (a)–(n) Input image at t_A, input image at t_B, ground truth, Ours, DGMAANet, ECTI, Diformer, BIFA, CD-STMamba, MSCANet, DAMFANet, RACDNet, RHighNet, and ChangeMamba.

contribution of frequency alignment rather than mere parameter count, we replaced the entire GFA module (M_1–M_3) with a standard multilayer perceptron (MLP) of equivalent parameter size. Disabling the GFA degrades the network to a spatial-only model, causing IoU to plummet by 3.51% on LEVIR-CD and 5.05% on CLCD. This significant performance gap, despite

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TABLE III
ABLATION STUDY OF FIVE COMPONENTS USING 11 CONFIGURATIONS ACROSS SIX DATASETS

Setting					LEVIR-CD		CLCD		BCDD		Google		DSIFN		SECOND	
M1	M2	M3	M4	M5	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
✓	×	✓	✓	✓	83.84	91.21	58.21	73.59	90.82	95.19	90.51	95.02	90.37	94.94	51.95	68.38
×	✓	✓	✓	✓	83.64	91.09	58.14	73.53	89.34	94.37	89.98	94.73	91.10	95.34	51.76	68.21
✓	✓	×	✓	✓	84.11	91.37	57.08	72.68	90.78	95.17	89.92	94.69	91.20	95.40	51.81	68.26
✓	✓	✓	×	✓	84.10	91.36	58.58	73.88	90.24	94.87	89.58	94.50	90.36	94.94	50.93	67.49
✓	✓	✓	✓	×	82.88	90.64	58.27	73.63	90.53	95.03	89.75	94.60	92.04	95.86	51.71	68.17
×	×	✓	✓	✓	82.17	90.21	56.74	72.40	87.24	93.19	88.63	93.97	89.80	94.63	48.71	65.51
×	✓	×	✓	✓	82.30	90.29	56.90	72.53	88.42	93.85	88.14	93.70	90.32	94.91	49.55	66.27
✓	×	×	✓	✓	81.91	90.06	55.77	71.61	89.08	94.22	88.95	94.15	89.78	94.61	48.67	65.47
✓	✓	✓	×	×	80.97	89.48	55.38	71.28	87.21	93.17	87.44	93.30	88.56	93.93	48.09	64.95
×	×	×	✓	✓	81.10	89.56	54.75	70.76	87.10	93.11	87.68	93.44	88.89	94.12	47.95	64.82
✓	✓	✓	✓	✓	84.61	91.66	59.80	74.84	91.22	95.41	90.68	95.11	92.47	96.09	53.07	69.34

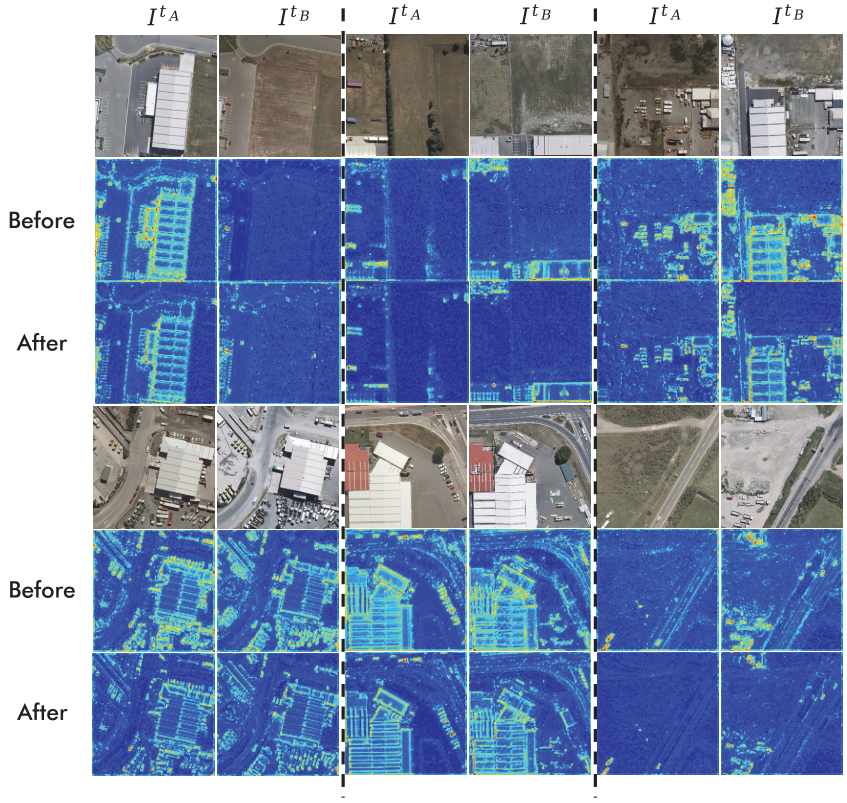


Fig. 14. Visual comparison of feature embeddings before and after the GFA module.

the identical parameter budget, validates the foundational and irreplaceable role of geometric-aware frequency disentanglement in robust CD. Furthermore, to provide a clear visual comparison, we visualize the feature embeddings when M_1, M_2, and M_3 are simultaneously eliminated (without-GFA).

Fig. 14 illustrates three representative input pairs, where the embeddings in the "after" row demonstrate a significantly clearer semantic understanding compared with those in the "before" row. Specifically, the boundaries of most objects, such as buildings and vehicles, exhibit a more distinct separation from the surrounding background.

Effectiveness of the BSFD (M4–M5): The BSFD resolves metric asymmetry and protects boundaries during temporal interaction. *SCSL (M4):* Removing SCSL drops IoU on the highly ambiguous SECOND dataset from 53.07% to 50.93%. This highlights that projecting features into a shared, neutral amplitude manifold is critical for eliminating directional metric bias. *SGAP (M5):* Without SGAP, the IoU on LEVIR-CD decreases by 1.73%. This indicates that forced style harmonization over-smooths genuine changes unless high-energy structural ridges are actively protected by phase priors. *BSFD Removal:* Similarly, to isolate the efficacy of our interaction mechanism from the impact of parameter scaling, we replaced the entire BSFD module (M_4–M_5) with a standard cross-attention mechanism of equivalent parameter size. Disabling the BSFD interaction mechanism leads to substantial performance degradation, with IoU dropping by 4.98% on SECOND and 3.64% on LEVIR-CD. These results underscore that symmetric, structure-protected alignment—rather than a generic attention-based fusion—is essential before computing feature differences to accurately distinguish genuine changes from seasonal noise.

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TABLE IV

COMPARISON OF DIFFERENT FREQUENCY-DOMAIN REPLACEMENTS FOR VERIFYING THE EFFECTIVENESS OF THE PROPOSED GFA MODULE. ALL VARIANTS USE THE SAME BACKBONE, DECODER, TRAINING PROTOCOL, AND INPUT RESOLUTION. ONLY THE FREQUENCY ALIGNMENT STRATEGY IS CHANGED. THE REPORTED METRICS ARE IOU AND $F1$

Frequency Alignment Strategy	LEVIR-CD		CLCD		BCDD		Google		DSIFN		SECOND	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
w/o GFA, spatial-domain baseline	81.10	89.56	54.75	70.76	87.10	93.11	87.68	93.44	88.89	94.12	47.95	64.82
Standard Cartesian Fourier filtering [62]	83.02	90.72	55.04	71.00	87.79	93.50	88.97	94.16	89.91	94.69	46.48	63.46
DCT-based frequency processing [63]	82.87	90.63	54.45	70.51	86.77	92.92	88.67	93.99	90.15	94.82	49.30	66.04
Learnable frequency filtering [64]	81.56	89.84	54.26	70.35	85.88	92.40	88.18	93.72	90.85	95.21	46.72	63.69
Proposed log-polar GFA	**84.61**	**91.66**	**59.80**	**74.84**	**91.22**	**95.41**	**90.68**	**95.11**	**92.47**	**96.09**	**53.07**	**69.34**

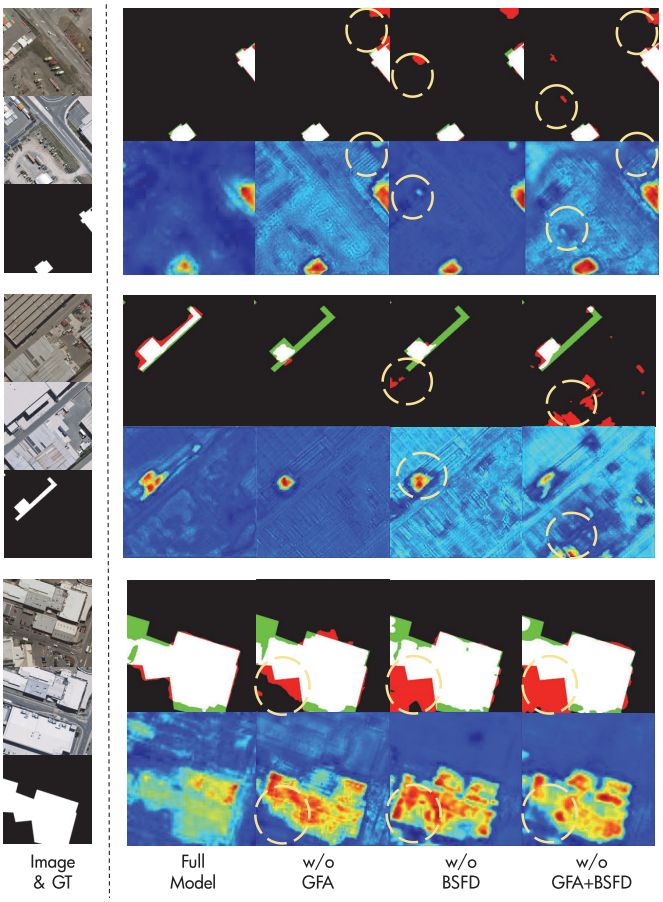


Fig. 15. Qualitative comparison of ablated variants and visualization of change-aware feature responses.

To provide a more comprehensive understanding of the network's internal mechanisms, Fig. 15 holistically visualizes the change-aware feature responses and the final prediction maps across different ablated variants. From a global perspective, these qualitative examples explicitly illustrate how the modules collaboratively guide the model's perception. As shown, the high-intensity response regions (i.e., the red and yellow areas) of the full model exhibit a highly consistent spatial alignment with the genuine change targets. For instance, in the first example, the pseudo-changes predicted by the three ablated variants directly correspond to their erroneously scattered high-intensity (yellow) feature activations in the background. Similar phenomena can also be consistently observed in the examples presented in the second and third rows.

F. Analysis of Frequency Alignment Strategy

To further verify whether the improvement comes from the proposed geometric alignment rather than a general frequency-domain enhancement, we compare GFA with several representative frequency representations in Table IV, including standard Cartesian Fourier filtering, DCT-based frequency processing, and learnable frequency filtering. Compared with the spatial-domain baseline, frequency-based variants generally provide improvements; for instance, standard Cartesian Fourier filtering increases the IoU on LEVIR-CD from 81.10% to 83.02%, indicating that frequency information is indeed useful for suppressing appearance-related disturbances. However, these alternatives are limited by fixed Cartesian sampling, global basis representations, or unconstrained filtering, which can even degrade performance in highly complex scenarios (e.g., the Cartesian method drops to 46.48% IoU on SECOND, underperforming the 47.95% spatial baseline). In contrast, the proposed log-polar GFA consistently achieves the best overall performance across all six datasets. It demonstrates particularly clear advantages on complex scenes that require precise geometric disentanglement, such as CLCD, BCDD, and SECOND. Specifically, GFA yields IoU scores of 59.80% on CLCD and 91.22% on BCDD, outperforming the best alternative frequency methods by substantial margins of 4.76% and 3.43%, respectively. On the challenging SECOND dataset, it reaches 53.07% IoU, exceeding the DCT-based method by 3.77%. This quantitatively demonstrates that explicitly aligning the receptive field with the radial distribution of spectral energy is significantly more effective than applying generic frequency enhancements.

G. Parameter Sensitivity and Module Analysis

In this section, we conduct a series of detailed investigations to evaluate the robustness and sensitivity of PDSS-Net. We first analyze the impact of different backbone architectures and scaling strategies on performance. Subsequently, we explore the influence of key hyperparameters and kernel configurations within the proposed SGAP and ACD modules to identify the optimal settings and validate their physical interpretability.

1) Analysis of Backbone Architectures: Table V demonstrates that scaling the feature extraction backbone does not yield monotonic improvements. Upgrading from ResNet-18 to

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TABLE V
QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT BACKBONES ACROSS SIX REMOTE SENSING DATASETS

Backbone	LEVIR-CD		CLCD		BCDD		Google		DSIFN		SECOND	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
ResNet-18	84.61	91.66	59.80	74.84	91.22	95.41	90.68	95.11	92.47	96.09	53.07	69.34
ResNet-34	**84.73**	**91.73**	59.90	74.92	91.57	95.60	**91.85**	**95.75**	92.13	95.90	**53.56**	**69.76**
ResNet-50	84.65	91.69	59.98	74.98	91.48	95.55	91.72	95.68	93.12	96.44	53.19	69.44
ResNet-101	84.33	91.50	**60.27**	**75.21**	**92.16**	**95.92**	91.51	95.57	92.52	96.11	53.06	69.33
ViT-B-16	84.28	91.47	59.95	74.96	91.25	95.42	90.87	95.22	92.40	96.05	52.84	69.14
ViT-L-16	84.30	91.48	59.69	74.76	90.99	95.28	91.18	95.39	**93.14**	**96.45**	52.68	69.01

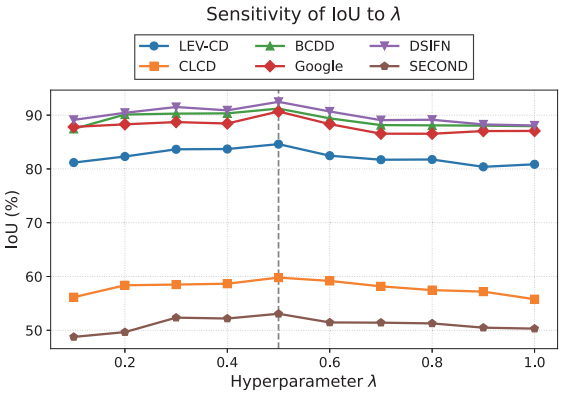


Fig. 16. Impact of the scaling factor λ on model performance across six datasets.

mid-sized networks like ResNet-34 brings marginal capacity gains; for instance, the IoU on the SECOND and Google datasets slightly increases from 53.07% and 90.68% to 53.56% and 91.85%, respectively. However, further expanding to the heavyweight ResNet-101 leads to performance degradation. Compared with ResNet-34, ResNet-101 experiences noticeable IoU drops of 0.40% on LEVIR-CD (from 84.73% to 84.33%) and 0.50% on SECOND (from 53.56% to 53.06%). This phenomenon highlights the scale discrepancy between remote sensing datasets and massive natural image datasets. With significantly fewer annotated samples, excessively deep networks easily overfit, memorizing specific radiometric noise or seasonal states rather than generalizing robustly.

Furthermore, the quantitative results explicitly expose the "data-hungry" limitation of Vision Transformers (ViTs) in the RSCD domain. Despite their massive parameter spaces, pure Transformer architectures like ViT-L-16 consistently underperform compared with CNNs. For example, on the complex CLCD dataset, ViT-L-16 yields an IoU of only 59.69%, failing to surpass any ResNet variants, while on the SECOND dataset, it trails the peak ResNet-34 performance by 0.88% in IoU (52.68% versus 53.56%). Lacking the inherent inductive biases of CNNs, ViTs struggle to optimize effectively on limited remote sensing data. Consequently, considering the minimal performance gap (e.g., only 0.12% IoU difference on LEVIR-CD) between ResNet-18 and larger variants, our PDSS-Net deliberately adopts the lightweight ResNet-18 to maximize computational efficiency, strictly prevent overfitting, and achieve the optimal capacity-data tradeoff.

2) Analysis of SGAP: In the SGAP module, the hyperparameter λ in (17) serves as a scaling factor that controls the protection intensity of the selective energy boosting for genuine structures. To investigate the impact of this parameter on the final model performance, we conducted a comprehensive evaluation across all six datasets with $\lambda \in [0.1, 1.0]$ at a step size of 0.1. Experimental results in Fig. 16 demonstrate that the IoU scores exhibit a consistent inverted U-shaped trend across all datasets, peaking precisely at $\lambda = 0.5$ (e.g., achieving an IoU of 84.61% on LEVIR-CD, 59.80% on CLCD, and a striking 92.47% on DSIFN).

Specifically, when the protection intensity is insufficient ($0.1 \leq \lambda \leq 0.4$), the structural protection barrier remains excessively weak, causing the model to over-rely on the baseline consensus amplitude $\mathcal{A}_{cons}$ during bitemporal interaction. As previously discussed, although $\mathcal{A}_{cons}$ effectively neutralizes seasonal differences, applying it blindly suppresses the high-frequency energy associated with genuine structural changes. Lacking adequate amplitude protection, true physical boundaries suffer from over-smoothing during style harmonization, preventing the model from precisely delineating the sharp edges of changed regions and resulting in suboptimal performance.

Conversely, when the protection intensity becomes excessive ($0.6 \leq \lambda \leq 1.0$), the model experiences a significant and continuous performance degradation (e.g., the IoU on CLCD drops from its peak to 55.78% at $\lambda = 1.0$, and LEVIR-CD declines to 80.86%). This decline occurs because an overly large scaling multiplier renders the energy boosting mechanism too aggressive. It not only amplifies high-energy topological features but also disrupts the unified consensus style that $\mathcal{A}_{cons}$ strives to maintain. In regions where the phase consistency gate $\mathcal{G}_{struct}$ yields imperfect estimations or severe illumination interference exists, an aggressive λ erroneously amplifies residual seasonal artifacts as structural features, thereby generating a substantial number of FPs. Ultimately, setting $\lambda = 0.5$ achieves an optimal dynamic balance between eliminating domain shifts caused by incoherent seasonal noise and actively shielding high-frequency topological features. Based on this empirical analysis, we fix λ to 0.5 in all implementation details of our network.

3) Analysis of ACD: We further investigate the sensitivity of the convolution kernel sizes $\mathcal{K}_{iso}$, $\mathcal{K}_{row}$, and $\mathcal{K}_{col}$ defined in (6). Several representative parameter combinations are evaluated, as summarized in Table VI. The corresponding performance across six distinct datasets is illustrated in Fig. 17.

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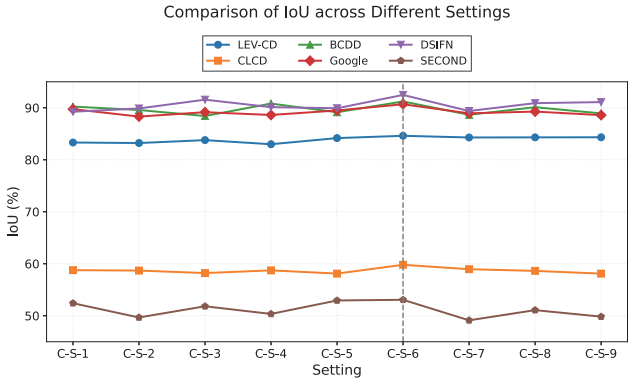


Fig. 17. Sensitivity analysis of different kernel size configurations ($\mathcal{K}_{iso}$, $\mathcal{K}_{row}$, and $\mathcal{K}_{col}$) in terms of the IoU metric across six CD datasets.

TABLE VI
DIFFERENT CONFIGURATIONS UNDER VARIOUS SETTINGS

Setting	$\mathcal{K}_{iso}$	$\mathcal{K}_{row}$	$\mathcal{K}_{col}$
C-S-1	1×1	1×9	9×1
C-S-2	2×2	1×9	9×1
C-S-3	3×3	1×9	9×1
C-S-4	1×1	1×7	7×1
C-S-5	2×2	1×7	7×1
C-S-6	3×3	1×7	7×1
C-S-7	1×1	1×5	5×1
C-S-8	2×2	1×5	5×1
C-S-9	3×3	1×5	5×1

The configuration C-S-6 (3×3 and 7×1 kernels) consistently achieves the highest IoU across all datasets, peaking at 91.22% on BCDD and 92.47% on DSIFN. This proves that the combination of a 3×3 spatial receptive field and a 7×1 structural kernel provides the optimal balance for feature extraction and alignment in RSCD.

Performance generally scales with kernel size: increasing the first-stage kernel from 1×1 to 3×3 yields consistent gains. Regarding the second stage, the 7×1 kernel outperforms both 5×1 (insufficient context) and 9×1 (excessive noise). This demonstrates that a moderate receptive field is essential for capturing structural details without introducing redundancy.

The model exhibits high stability on datasets like LEVIR-CD (fluctuation <2%) while showing greater sensitivity on complex datasets like SECOND. The fact that C-S-6 remains the superior setting across all six datasets, despite their varying characteristics, validates the strong generalization and robustness of the PDSS-Net architecture.

H. Frequency-Domain Structural Fidelity Analysis

To further validate our network's superiority in preserving geometric topology, we visualize the FFT magnitude spectra of the predicted change masks in Fig. 18.

The ground-truth masks (c) exhibit distinct, sharp orthogonal energy ridges in the frequency domain (d), corresponding to the precise physical boundaries of the changed buildings. It is evident that the FFT magnitudes of our PDSS-Net (e) highly resemble the ground truth, successfully reconstructing these critical high-frequency structural ridges. In

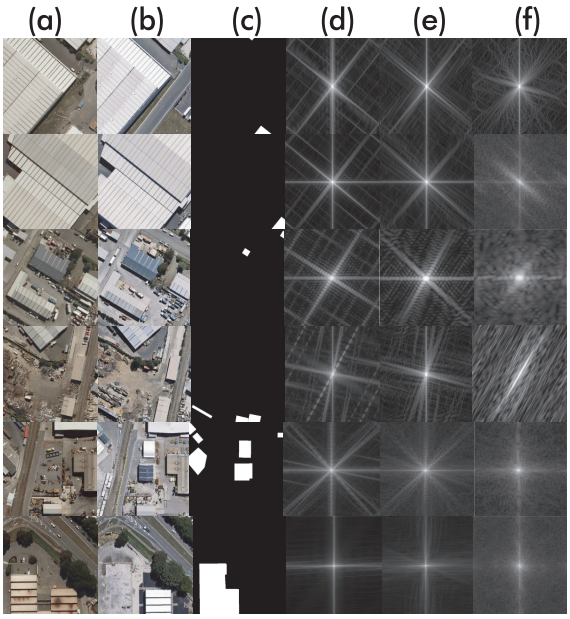


Fig. 18. (a)–(f) Input image at t_A, input image at t_B, ground truth, FFT magnitudes of ground truth, FFT magnitudes of Ours, and FFT magnitudes of RACDNet.

contrast, the frequency spectra produced by RACDNet (f) suffer from severe blurring and energy diffusion, indicating a loss of high-frequency edge information. This frequency-domain comparison explicitly proves that our method not only locates changes accurately but also maintains high fidelity to the genuine structural boundaries.

I. Resistance to Nonstationary Environmental Noise

To evaluate the robustness of PDSS-Net to image degradation, we analyze the additive Gaussian noise experiments summarized in Table VII. Noise is added to the second-temporal image (t_B) at standard deviations of $\sigma = 0.10$ and $\sigma = 0.20$. The clean-input results in the same table provide the reference for measuring performance degradation across the six datasets.

As reported in Table VII, PDSS-Net achieves IoU scores of 82.28%, 57.25%, 88.69%, 87.93%, 90.18%, and 50.73% on LEVIR-CD, CLCD, BCDD, Google, DSIFN, and SECOND, respectively, at $\sigma = 0.10$. Relative to the clean-input results, these values correspond to decreases of 2.33, 2.55, 2.53, 2.75, 2.29, and 2.34 percentage points, respectively. Thus, at this tested noise level, the IoU reduction remains below 3 percentage points on each dataset.

Increasing the noise level to $\sigma = 0.20$ further reduces the IoU to 79.19%, 54.75%, 85.36%, 84.12%, 86.91%, and 48.23% on LEVIR-CD, CLCD, BCDD, Google, DSIFN, and SECOND, respectively. For example, the IoU on LEVIR-CD decreases from 84.61% under clean conditions to 82.28% at $\sigma = 0.10$ and 79.19% at $\sigma = 0.20$, corresponding to reductions of 2.33 and 5.42 percentage points from the clean reference. All six datasets show lower IoU at the higher tested noise level. These results quantify the performance degradation of

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TABLE VII

CONTROLLED PERTURBATION ANALYSIS ON SIX RSCD DATASETS. THE EVALUATED PERTURBATIONS INCLUDE MULTIPLICATIVE AMPLITUDE SCALING, BOUNDED PHASE NOISE, CLOUD OCCLUSION, SINGLE-TEMPORAL RESOLUTION DEGRADATION, GAUSSIAN BLUR, AND ADDITIVE GAUSSIAN NOISE. IOU AND $F1$-SCORE ARE REPORTED IN PERCENTAGE (%)

Perturbation Type	Perturbation Strength	LEVIR-CD		CLCD		BCDD		Google		DSIFN		SECOND	
		IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
Clean	–	84.61	91.66	59.80	74.84	91.22	95.41	90.68	95.11	92.47	96.09	53.07	69.34
Amplitude	$\alpha = 5\%$	83.92	91.26	58.16	73.55	89.35	94.38	87.60	93.39	91.06	95.32	50.35	66.98
	$\alpha = 10\%$	83.45	90.98	58.07	73.47	88.98	94.17	86.42	92.72	90.01	94.74	48.56	65.37
	$\alpha = 20\%$	78.79	88.14	55.51	71.39	86.93	93.01	84.74	91.74	89.34	94.37	46.90	63.85
	$\alpha = 30\%$	77.39	87.25	55.16	71.10	79.28	88.44	80.38	89.12	79.40	88.52	40.44	57.59
Phase	$\beta = \pi/64$	80.09	88.94	57.89	73.33	88.83	94.08	87.13	93.12	90.57	95.05	49.90	66.58
	$\beta = \pi/32$	79.11	88.34	56.60	72.29	87.31	93.23	86.11	92.54	88.61	93.96	47.92	64.79
	$\beta = \pi/16$	77.46	87.30	55.80	71.63	86.22	92.60	82.59	90.46	88.15	93.70	45.97	62.99
	$\beta = \pi/8$	77.33	87.22	54.28	70.37	76.30	86.56	77.12	87.08	78.55	87.99	38.69	55.79
Cloud Occlusion	$r_{\text{cloud}} = 5\%$	82.61	90.48	57.98	73.40	89.46	94.44	88.56	93.93	90.74	95.15	51.43	67.93
	$r_{\text{cloud}} = 10\%$	81.75	89.96	57.04	72.64	88.56	93.93	87.48	93.32	89.91	94.69	50.52	67.13
	$r_{\text{cloud}} = 15\%$	80.76	89.36	56.05	71.84	87.54	93.36	86.04	92.50	88.81	94.07	49.61	66.32
	$r_{\text{cloud}} = 20\%$	77.42	87.27	53.49	69.70	84.17	91.40	82.51	90.42	85.44	92.15	47.07	64.01
Resolution Degradation	$s = 2\times$	82.46	90.39	57.58	73.08	88.83	94.08	88.11	93.68	90.30	94.90	51.06	67.60
	$s = 3\times$	80.88	89.43	56.19	71.95	87.09	93.10	86.28	92.63	88.26	93.76	49.77	66.46
	$s = 4\times$	76.57	86.73	52.67	69.00	82.26	90.27	80.86	89.42	83.75	91.16	46.31	63.30
Gaussian Blur	$\sigma = 2$	81.78	89.98	56.96	72.58	87.88	93.55	86.89	92.99	89.61	94.52	50.38	67.00
	$\sigma = 4$	78.29	87.82	53.88	70.03	83.97	91.29	82.30	90.29	85.86	92.39	47.40	64.31
Gaussian Noise	$\sigma = 0.10$	82.28	90.28	57.25	72.81	88.69	94.01	87.93	93.58	90.18	94.84	50.73	67.31
	$\sigma = 0.20$	79.19	88.39	54.75	70.76	85.36	92.10	84.12	91.38	86.91	93.00	48.23	65.07

PDSS-Net under additive Gaussian noise and complement the other controlled perturbation experiments discussed below.

J. Controlled Perturbation and Robustness Analysis

To comprehensively evaluate the robustness of PDSS-Net under representative cross-temporal disturbances, we conduct controlled perturbation experiments covering amplitude variation, phase noise, cloud occlusion, single-temporal resolution degradation, Gaussian blur, and Gaussian noise, as summarized in Table VII. For amplitude perturbation, the Fourier amplitude of one temporal image is multiplicatively scaled by a bounded factor controlled by α, thereby simulating radiometric and seasonal appearance variations. Phase perturbation adds bounded phase noise with magnitude β to disrupt spatial structure and boundary information. Cloud occlusion is introduced only into the second-temporal image using irregular synthetic cloud-like masks, where r_{cloud} denotes the proportion of cloud-covered pixels. For resolution degradation, the first-temporal image is downsampled by a factor s and then restored to its original size using bilinear interpolation, while the second-temporal image remains unchanged. Gaussian blur with standard deviation σ suppresses local edges and fine spatial details, whereas additive Gaussian noise with standard deviation σ simulates random sensor-level intensity corruption.

As shown in Table VII, the performance decreases consistently as the perturbation strength increases, indicating that the controlled settings produce a meaningful progression from mild to severe cross-temporal inconsistency. Under mild amplitude perturbation ($\alpha = 5\%$), the IoU decreases by only 0.69, 1.64, 1.87, 3.08, 1.41, and 2.72 percentage points on LEVIR-CD, CLCD, BCDD, Google, DSIFN, and SECOND, respectively. Even at $\alpha = 20\%$, PDSS-Net retains IoUs of 78.79%, 55.51%, 86.93%, 84.74%, 89.34%, and 46.90% on the six datasets. More evident degradation appears at $\alpha = 30\%$, particularly on BCDD, DSIFN, and SECOND, where the IoU falls to 79.28%, 79.40%, and 40.44%, respectively. This pattern suggests that moderate amplitude changes mainly alter radiometric appearance, whereas large amplitude scaling increasingly distorts the intertemporal appearance correspondence required for reliable change discrimination.

Phase perturbation causes a generally stronger performance reduction than amplitude perturbation at comparable mild settings, which is consistent with the greater importance of phase information for preserving object structure, spatial layout, and boundary localization. For example, the IoU on LEVIR-CD decreases from 84.61% to 80.09% at $\beta = \pi/64$, while the corresponding value under $\alpha = 5\%$ remains 83.92%. Similar behavior can be observed on CLCD, BCDD, Google, DSIFN, and SECOND, where $\beta = \pi/64$ yields IoUs of 57.89%, 88.83%, 87.13%, 90.57%, and 49.90%, respectively. When the phase disturbance increases to $\beta = \pi/8$, the IoU decreases to 77.33% on LEVIR-CD, 54.28% on CLCD, 76.30% on BCDD, 77.12% on Google, 78.55% on DSIFN, and 38.69% on SECOND. The larger decline on datasets such as BCDD, Google, DSIFN, and SECOND demonstrates that severe phase corruption directly weakens structural consistency and makes pseudo-changes more difficult to distinguish from genuine land-cover changes.

The spatial and pixel-domain perturbations show similarly coherent trends. For cloud occlusion, increasing r_{cloud} from 5% to 20% reduces the IoU from 82.61% to 77.42% on LEVIR-CD, from 57.98% to 53.49% on CLCD, and from 89.46% to 84.17% on BCDD. The corresponding values on Google, DSIFN, and SECOND decrease from 88.56%, 90.74%, and 51.43% to 82.51%, 85.44%, and 47.07%, respectively. These results indicate that the model remains relatively stable under limited occlusion, while larger missing regions inevitably remove discriminative temporal evidence. Resolution degradation also causes progressive loss of fine spatial details. At $s = 2\times$, the IoU remains above 88% on BCDD, Google, and DSIFN, whereas at $s = 4\times$ it drops to 82.26%, 80.86%, and

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TABLE VIII

ABLATION STUDY OF REPRESENTATIVE LOSS FUNCTIONS. ALL VARIANTS USE THE SAME FULL PDSS-NET ARCHITECTURE, TRAINING PROTOCOL, INPUT RESOLUTION, AND EVALUATION METRICS. ONLY THE LOSS FUNCTION IS CHANGED. THE REPORTED METRICS ARE IOU AND $F1$

Loss Function	LEVIR-CD		CLCD		BCDD		Google		DSIFN		SECOND	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
BCE	**84.61**	**91.66**	**59.80**	**74.84**	91.22	95.41	**90.68**	**95.11**	92.47	96.09	53.07	69.34
Dice	79.39	88.51	54.50	70.55	86.43	92.72	85.62	92.25	88.14	93.70	49.70	66.40
BCE + Dice	84.02	91.32	58.31	73.67	90.91	95.24	89.43	94.42	**92.73**	**96.23**	**53.62**	**69.81**
Focal Loss	84.35	91.51	58.75	74.02	**91.36**	**95.48**	89.73	94.59	91.85	95.75	50.56	67.16
Boundary Loss	83.63	91.09	59.13	74.32	90.42	94.97	90.03	94.75	92.26	95.97	50.21	66.85

83.75%, respectively. LEVIR-CD and CLCD show comparable declines from 82.46% and 57.58% to 76.57% and 52.67%, confirming that severe single-temporal resolution mismatch mainly damages small objects and accurate boundary recovery.

Gaussian blur and Gaussian noise further verify the sensitivity of CD to local-detail suppression and random intensity corruption. Increasing the blur strength from $\sigma = 2$ to $\sigma = 4$ lowers the IoU from 81.78% to 78.29% on LEVIR-CD, from 87.88% to 83.97% on BCDD, from 86.89% to 82.30% on Google, and from 89.61% to 85.86% on DSIFN. Likewise, increasing the Gaussian noise level from $\sigma = 0.10$ to $\sigma = 0.20$ reduces the IoU from 82.28% to 79.19% on LEVIR-CD and from 88.69%, 87.93%, and 90.18% to 85.36%, 84.12%, and 86.91% on BCDD, Google, and DSIFN, respectively. SECOND exhibits lower absolute accuracy under all perturbations, which is consistent with its more challenging and heterogeneous change patterns, but still shows a gradual rather than abrupt degradation. Overall, PDSS-Net maintains strong performance under mild and moderate disturbances, while the systematic decline under severe settings confirms that the proposed tests effectively cover complementary sources of cross-temporal inconsistency. The relatively larger sensitivity to phase corruption and strong spatial-detail degradation also supports the structural motivation of the proposed phase-dominant sensing and symmetric interaction design.

K. Loss Function Analysis

We further compare several representative loss functions to justify the use of BCE in the proposed framework. As detailed in Table VIII, BCE achieves the most robust and consistent performance across the majority of datasets. While combining BCE with Dice loss yields marginal improvements on certain datasets—such as increasing the IoU from 92.47% to 92.73% on DSIFN and from 53.07% to 53.62% on SECOND—it simultaneously causes performance degradation elsewhere, dropping the IoU on LEVIR-CD from 84.61% to 84.02%. Relying solely on Dice loss leads to severe degradation, with the IoU plummeting to 79.39% on LEVIR-CD and 49.70% on SECOND. This occurs because Dice loss emphasizes global region overlap, which inherently weakens pixel-level supervision for small or fragmented changes. Similarly, while Focal loss slightly improves the IoU on the BCDD dataset to 91.36% (compared with BCE's 91.22%), neither Focal nor Boundary loss provides consistent gains across all six benchmarks. These quantitative comparisons explicitly indicate that the state-of-the-art performance of PDSS-Net stems fundamentally from the proposed frequency-domain disentanglement and symmetric bitemporal interaction, rather than relying on complex loss formulations. Consequently, BCE is adopted as a simple, stable, and highly effective objective for fair comparison.

TABLE IX

EFFICIENCY COMPARISON OF DIFFERENT METHODS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	Year	Params (M)	FLOPs (G)	Train FPS	Test FPS
FCEF [5]	2018	**1.35**	**12.44**	83.22	311.02
FCSiam [5]	2018	**1.35**	**12.44**	**87.77**	**343.48**
MSCANet [54]	2022	17.11	58.76	23.82	69.75
DGMAANet [25]	2024	37.62	72.27	16.23	61.05
DAMFANet [55]	2024	17.06	45.47	22.97	86.00
ECTI [56]	2024	19.32	41.52	31.77	126.05
Diformer [13]	2024	21.17	44.98	28.32	102.63
BIFA [3]	2024	28.10	62.49	21.35	76.12
STADECDNet [20]	2024	11.85	103.83	33.97	125.47
ELGCNet [12]	2024	10.57	494.35	5.43	19.03
Changemamba [19]	2024	32.61	80.03	4.71	17.03
DSFDCD [11]	2025	9.15	69.66	19.51	52.31
RHighNet [24]	2025	120.79	69.47	3.43	9.04
STRobustNet [53]	2025	5.23	13.21	34.97	133.52
ISDANet [57]	2025	6.81	62.91	3.11	17.17
CD-STMamba [18]	2025	64.31	268.34	23.90	94.11
RACDNet [14]	2026	24.67	34.36	3.80	12.28
WHFCE [15]	2026	21.33	401.21	4.68	16.97
Ours (PDSS-Net)	2026	23.08	33.76	27.28	114.10

TABLE X

CROSS-DATASET GENERALIZATION COMPARISON. ALL MODELS ARE TRAINED ON THE SOURCE DATASET AND DIRECTLY EVALUATED ON THE TARGET DATASET WITHOUT FINE-TUNING. THE REPORTED METRICS ARE IOU AND $F1$

Source Dataset	Target Dataset	DGMAANet		Ours	
		IoU	F1	IoU	F1
LEVIR-CD	Google	71.51	83.39	**77.78**	**87.50**
LEVIR-CD	BCDD	66.26	79.71	**74.78**	**85.57**
BCDD	LEVIR-CD	58.57	73.87	**62.87**	**77.20**
BCDD	Google	68.87	81.57	**75.13**	**85.80**
Google	BCDD	60.88	75.68	**72.71**	**84.20**
Google	LEVIR-CD	56.42	72.14	**64.94**	**78.75**

L. Efficiency Comparison

In practical deployments of remote sensing image CD, models must strike a good balance among detection accuracy, parameter count (Params), and floating-point operations (FLOPs). To provide a more complete runtime comparison, we additionally report training speed (Train FPS) and inference

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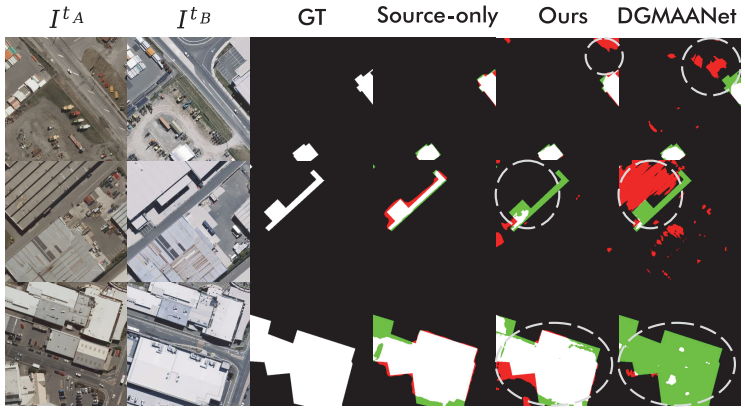


Fig. 19. Qualitative visualization of cross-dataset generalization under different source-to-target settings. All models are trained on the source dataset and directly evaluated on the unseen target dataset without fine-tuning.

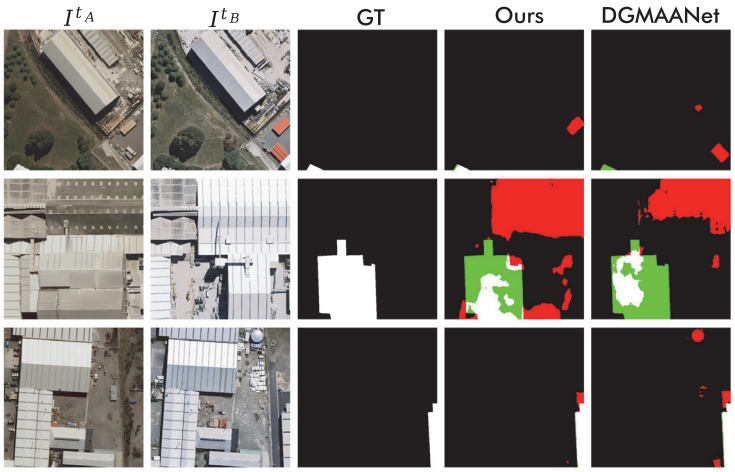


Fig. 20. Representative failure cases of the proposed method.

speed (Test FPS) in Table IX. Both are measured as processed image pairs per second on a single NVIDIA RTX 4090 GPU using paired 512 $\times$ 512 RGB inputs and a batch size of 1. Train FPS includes the forward pass, loss computation, backward pass, and optimizer update, whereas Test FPS denotes synchronized forward inference after warm-up; data loading is excluded. By integrating the mIoU performance of various models across six datasets, we conducted a comprehensive efficiency analysis of the proposed PDSS-Net in Table IX.

First, the proposed PDSS-Net features a parameter count of 23.08M and a computational cost of 33.76G FLOPs. Despite this lightweight architecture, PDSS-Net achieved the highest average mIoU of 78.64% across the six public datasets. This demonstrates that physics-aware structural feature extraction in the frequency domain (e.g., the GFA and BSFD modules) can precisely capture change semantics at a minimal computational cost, effectively avoiding redundant feature stacking. Compared with RACDNet, also proposed in 2026, which has 24.67M parameters and 34.36G FLOPs, PDSS-Net achieves a slightly lower computational overhead while its average mIoU (78.64%) significantly outperforms RACDNet's 72.68%. This indicates that under comparable computational constraints, our method exhibits significantly superior feature decoupling and anti-interference capabilities. Regarding runtime efficiency, PDSS-Net reaches 27.28 Train FPS and 114.10 Test FPS. Although the extremely lightweight FCEF and FCSiam models achieve higher speeds, PDSS-Net provides substantially stronger detection accuracy while maintaining practical training and inference throughput. It also runs considerably faster than several recent complex models, including DGMAANet, Changemamba, RHighNet, ISDANet, RACDNet, and WHFCE, demonstrating a favorable balance between accuracy and runtime efficiency.

Second, several recent advanced models have sought to improve performance by introducing complex attention mechanisms or excessively large receptive fields, leading to a surge in computational complexity. For instance, the WHFCE and the ELGCNet consume up to 401.21G and 494.35G FLOPs, respectively. Similarly, the Mamba-based CD-STMamba requires a computational load of 268.34G FLOPs. In contrast, PDSS-Net's computational cost is only about 1/12 of WHFCE's and 1/8 of CD-STMamba's, yet it surpasses CD-STMamba's average mIoU (74.12%) by more than 4.5 percentage points. This highlights PDSS-Net's substantial deployment potential on resource-constrained computing platforms.

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In terms of parameter count, RHighNet contains a staggering 120.79M parameters, and DGMAANet utilizes 37.62M parameters. However, RHighNet achieves an average mIoU of only 72.20%. This further validates the observation made in our ablation study: given the limited availability of remote sensing data, blindly increasing network depth and parameter scale easily leads to overfitting. With less than one-fifth of RHighNet's parameters, PDSS-Net (23.08M) achieves an accuracy improvement of over 6.4%.

M. Cross-Dataset Generalization Analysis

To further evaluate the domain generalization ability of PDSS-Net, we conduct cross-dataset experiments, where the model is trained on one source dataset and directly evaluated on another unseen target dataset without fine-tuning. As reported in Table X, PDSS-Net consistently outperforms DGMAANet across all six source-to-target settings. On average, PDSS-Net improves the IoU from 63.75% to 71.37% and the $F1$-score from 77.73% to 83.17%. This indicates that the proposed frequency-domain disentanglement is not limited to in-domain supervised fitting, but can better transfer to unseen geographic scenes and imaging conditions.

Fig. 19 further provides qualitative evidence. Under cross-dataset evaluation, the target-domain images usually exhibit obvious differences in building density, background texture, illumination, and spatial resolution. Conventional spatial modeling is more likely to activate unchanged regions as pseudo-changes under such domain shifts. In contrast, PDSS-Net produces cleaner prediction maps in unseen target domains. This can be attributed to the combination of GFA and BSFD: GFA extracts more stable structural representations by reducing amplitude-dominated appearance interference, while BSFD compares bitemporal features in a shared consensus amplitude space to reduce asymmetric interaction errors.

N. Failure-Case Analysis and Reproducibility

Although PDSS-Net achieves competitive performance, several failure cases remain, as shown in Fig. 20. The first type of error occurs around object boundaries, especially for large buildings in the Google dataset. In these cases, the model can locate the changed object, but the predicted mask may slightly deviate from the ground-truth contour, resulting in narrow FP or FN regions. This is mainly related to spatial resolution, roof shadows, and annotation uncertainty.

The second type of error appears in subtle farmland changes in CLCD. Such regions usually have weak structural contrast and irregular boundaries, making them difficult to distinguish from seasonal vegetation variations. Although GFA suppresses many appearance-induced pseudo-changes, farmland changes with weak phase discontinuities may still be partially missed. The third type involves very small or fragmented changed objects. These targets occupy only a few pixels and may be weakened during multiscale feature aggregation, leading to incomplete detection. Finally, SGAP may occasionally amplify high-frequency artifacts when background textures, shadows, or annotation noise also present strong phase responses. This explains the remaining isolated FPs in some complex scenes.

These observations indicate that the proposed method improves robustness against dominant pseudo-changes, but boundary uncertainty, extremely subtle changes, and very small fragmented objects remain challenging.

V. Conclusion

This article proposes the PDSS-Net to tackle phenological interference in bitemporal CD via frequency-domain disentanglement. Specifically, the GFA module rectifies spectral geometric bias to extract robust monotemporal features, while the BSFD module eliminates interaction asymmetry using a symmetric consensus mechanism. Extensive experiments on six benchmarks demonstrate that PDSS-Net achieves state-of-the-art accuracy and exceptional noise robustness with a highly efficient computational footprint.

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